



2025 FRM Part II

百题巅峰班

操作风险及弹性

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3. Operational Risk and Resiliency

3.1. Holistic Overview of Operational Risk and Resilience

3.1.1. 重要知识点

3.1.1.1. Operational Risk Management

- The risk of loss resulting from inadequate or failed internal processes, people and systems or from external events. The definition includes legal risk but excludes strategic and reputation risk.
- Basel II classifies loss events into seven categories: Internal Fraud; External Fraud; Employment Practices and Workplace Safety; Clients, Products, and Business Practices; Damage to Physical Assets; Business Disruption and System Failures; Execution, Delivery, and Process Management
- Three Lines of Defense: Business Line Management; Functionally Independent Corporate Operational Risk Function (CORF); Independent Review and Challenge of the bank's operational risk management controls, processes and systems.
- **Role of Board of Directors:** establish, approve and periodically review the operational risk management framework
- **Role of Senior Management:** develop a clear, effective and robust governance structure with well defined, transparent and consistent lines of responsibility; consistently implementing and maintaining throughout the organization policies, processes and systems for managing operational risk in all of the bank's material products, activities, processes and systems consistent with the risk appetite and tolerance.

3.1.2. 基础题

- Q-1.** Suppose a broker-dealer has a loss that occurs from a failure in properly processing and settling a transaction. According to Basel II operational risk categories, this type of event loss would be categorized as:
- A. Business Disruption and System Failures.
 - B. Clients, Products, and Business Practices.
 - C. Execution, Delivery, and Process Management.

D. Employment Practices and Workplace Safety

Q-2. A large investment bank has just acquired a smaller regional competitor and is extending its best practices in the field of operational risk to the newly acquired company. As part of this process, management of the new subsidiary is reviewing which responsibilities should be assumed by the board of directors and which should be assumed by senior management. For which of the following should the board of directors be responsible?

- A. Implementing operational risk management systems across the organization
- B. Develop a clear, effective and robust governance structure
- C. Assigning responsibilities to, and reporting relationships between, the bank's risk managers
- D. Periodically reviewing and approving the operational risk management framework

Q-3. The CEO of a large bank has reported that the bank's framework for managing operational risk are consistent with Basel II and Basel III model for operation risk governance. Which of the following actions and principles of the bank is correct?

- A. The bank considers identification and management of risk as the second line of defense.
- B. The bank considers independent review and audit of the risk processes and systems as the third line of defense.
- C. The bank includes damaged reputation due to a failed merger in its measurement of operational risk.
- D. The bank excludes destruction by fire or other external catastrophes from its measurement of operational risk.

Q-4. In its efforts to enhance its enterprise risk management function. Countryside Bank introduced a new decision-making process based on economic capital that involves assessing sources of risk across different business units and organizational levels. Which of the following statements regarding the correlations between these risks is correct?

- A. Correlations between the risks in the asset and liability sides of the balance sheet can be

changed by management decisions.

- B. Generally, correlations between broad risk types such as credit, market, and operational risk are well understood and are easy to estimate at the individual firm level.
- C. Correlations between business units are only relevant in deciding total firm-wide economic capital levels and are not relevant for decisions at the individual business unit or project level.
- D. The introduction of correlations into firm-wide risk evaluation will result in a total VaR that, in general, is greater than or equal to the sum of individual business unit VaRs.

Q-5. While building the bank's enterprise risk management system, a risk analyst takes an inventory of firm risks and categorizes these risks as market, credit, or operational. Which of the following observations of the bank's data should be considered unexpected if compared to similar industry data?

- A. The operational risk loss distribution has a large number of small losses and therefore, a relatively low mode.
- B. The operational risk loss distribution is symmetric and fat-tailed.
- C. The credit risk distribution is asymmetric and fat-tailed.
- D. The market risk distribution is similar to the distribution of the return on a portfolio of securities.

Q-6. Each of the following is a key issue related to external datasets in modeling operational losses except:

- A. The lack of any vendors with centralized database necessitates the manual collection of peer data.
- B. The relevance question of whether past loss events at other institutions seem likely or even plausible for the user bank going forward.
- C. The quantity question: there may not be enough data points in the external dataset that can supplement the internal loss data.
- D. The under-reporting problem that not all loss events reach the public domain.

- Q-7.** A UK-based retail brokerage firm has recently experienced rapid growth through a series of acquisitions and plans to improve its operational resilience in order to comply with new requirements issued by national regulators. The CRO asks an operational risk manager to assess best practices in this area and to suggest potential actions that the firm should take to meet this objective. Which of the following actions would be most appropriate for the manager to recommend in order to comply with the regulatory guidelines concerning operational resilience for important business services?
- A. Increase the existing performance incentives for the firm's brokers and sales representatives to further enhance their productivity.
 - B. Reduce costs by consolidating the human resources and payroll functions of the acquired firms into a single firm-wide process.
 - C. Reserve additional economic capital for operational risk to provide an incremental capital cushion against potentially extreme operational losses.
 - D. Increase the bandwidth capacity for the firm's equity trading platform and procure remote backup capabilities so the platform can continue to function during an outage.
- Q-8.** The board of directors of a midsize bank has recommended that the bank improve its processes for managing operational risk. The CEO asks an enterprise risk manager to review the bank's tools and processes for managing operational risk and to suggest improvements that are consistent with Basel II and Basel III guidelines for operational risk governance. The bank also plans to adopt the new Basel III standardized approach (SA) to determine its regulatory capital for operational risk. Which of the following actions should the manager recommend that the bank take?
- A. Use third-party outsourcing agreements to replace most internal controls performed by senior managers and business line managers.
 - B. Develop an internal approach to model the distribution of operational risk losses and use it to determine the bank's regulatory capital.
 - C. Require an independent review of the bank's operational risk management framework.
 - D. Designate the risk management function as the primary owner of risk exposures within

each business line.

- Q-9.** A bank owned five retail branch buildings that were destroyed in a hurricane. Several months later, the bank received a settlement payment from its insurance company that covered a portion of the value of the buildings. A risk manager is preparing an operational risk incident report of this loss for the national regulator and wants to ensure that all the appropriate costs are included in the report and that the reporting process meets best industry practices. Which of the following statements correctly describes how the losses from this event should be reported?
- A. The report should include the gross loss incurred as a result of the event, but not the settlement payment received from the insurance company.
 - B. The reported loss should include an estimate of the opportunity costs of banking business lost at the affected branches.
 - C. The reported loss should include the legal costs paid to obtain construction permits to rebuild the destroyed branch buildings.
 - D. The losses should be reported as a market risk event rather than an operational risk event because they resulted in a reduction of the value of the bank's real estate portfolio.
- Q-10.** An operational risk manager is presenting to a group of representatives from financial institutions about the advantages and disadvantages of different approaches that firms use to manage operational risk. The manager provides an example of a financial institution that uses the Swiss cheese model as a control structure to manage the risk of operational losses due to incorrect transaction processing. Which of the following correctly describes an expected impact of the firm's use of the Swiss cheese model?
- A. Controls are structured in a way that a failure of a single key control would directly result in an operational loss.
 - B. Controls are structured in a way that every layer of controls would have to fail in order for an operational loss to occur.
 - C. When using this model, the probability of each subsequent control failure is dependent

on whether a previous control for the same process failed.

- D. By using this model, the probability of an extreme operational loss decreases significantly but the probability of small operational losses increases.

Q-11. An enterprise risk manager at a large bank is reviewing the results of a recently completed firm-wide risk control self-assessment (RCSA). The RCSA indicates that there are multiple classes of risk for which the bank should improve its risk controls, and the manager believes the firm should consider transferring some of these risks. The manager identifies the types of risks the bank should transfer, and also assesses the impact of different risk transfer strategies on the bank's risk profile and its regulatory capital calculation. Which of the following actions should the manager recommend for the bank to take?

- A. Use a captive insurance subsidiary to cover the bank's tail risk exposure.
- B. Increase the bank's insurance coverage in order to benefit by deducting the cost of the premiums from the bank's required Basel operational risk capital.
- C. Acquire insurance against cyber risks and business interruptions from an insurance company.
- D. Transfer credit risk and fraud risk by outsourcing core operations such as loan pricing and review of new account applications.

Q-12. A community bank uses the three lines of defense approach to manage its operational risk exposures. The bank is in the process of implementing an enterprise risk management (ERM) framework, and the CRO decides to extend the three lines of defense approach to its implementation of ERM. The CRO also wants to ensure that the ERM framework appropriately reflects the bank's risk appetite and risk culture. Which of the following actions should the CRO recommend for the bank to take?

- A. The third line of defense should continuously monitor the bank's implementation of its ERM framework to ensure its effectiveness.
- B. Business line managers, as part of the first line of defense, should have the authority to

take on risk exposures within the bank's risk appetite limits.

- C. The bank should implement a set of risk culture indicators as part of its ERM framework in order to accurately quantify the losses that could occur due to failures of risk culture.
- D. As part of the second line of defense, the executive committee should perform an independent review of the bank's risk management framework.

Q-13. An internal auditor at a national bank is reviewing the assumptions of the scenarios that are used in the bank's stress tests. The auditor wants to ensure that the different sets of assumptions are plausible and that they exhibit the property of coherence. Which of the following combinations of assumptions used in a stress scenario would provide the auditor the greatest cause for concern about their coherence?

- A. The USD, EUR, and JPY all depreciate at the same time against a basket of world currencies.
- B. Correlations between returns of commodities used for industrial production and returns of industrial sector equities increase at the same time.
- C. Yields on BBB-rated corporate debt increase as yields on AAA-rated short-term sovereign debt decrease.
- D. Monetary authorities respond to higher consumer prices by raising central bank lending rates.

Q-14. An internal auditor at a large bank is reviewing the bank's economic capital framework to ensure that it meets best practices. The auditor identifies deficiencies in the bank's governance framework as well as the process used to determine the firm-wide economic capital and asks the CRO to suggest corrective actions that conform with best practices. Which of the following actions should the CRO recommend?

- A. Require business unit managers to challenge the assumptions for their unit's capital model before providing final approval.
- B. Calculate the bank's aggregate economic capital by summing its exposures for different risk types.

- C. Incorporate a set of escalation procedures into the bank's contingency plan for its economic capital policy.
 - D. Discourage the use of macroeconomic scenarios developed by third-party vendors to stress test economic capital models.
- Q-15.** A large bank is reviewing its processes and procedures to manage operational risk in accordance with best practices established by the Basel Committee. In implementing the three lines of defense model, which of the following statements is correct?
- A. The internal audit function should serve as the first line of defense and continually validate operational procedures used by the business lines.
 - B. Business line managers, as part of the first line of defense, should provide a credible challenge to the internal audit function.
 - C. The corporate operational risk function, as part of the second line of defense, should challenge risk inputs from business line managers.
 - D. The corporate operational risk function should serve as the third line of defense and validate model assumptions made by senior management.
- Q-16.** A senior risk consultant is discussing different types of operational risk with a group of managers at Bank JKY. During the discussion, the consultant describes the seven Basel event categories for the classification of operational losses and provides examples of losses in each of the categories. Which of the following describes an operational loss that Bank JKY should classify in the Execution, Delivery, and Process Management loss category?
- A. A client files a lawsuit against Bank JKY for a breach of fiduciary responsibility in the client's investment account, and Bank JKY pays a monetary settlement to the client.
 - B. The data capture system of Bank JKY fails to capture correct market prices, causing OTC derivative trades to occur at incorrect prices and resulting in significant losses.
 - C. A rogue trader makes a series of unauthorized derivative trades, which are liquidated by Bank JKY for a sizable loss after discovered.

- D. A corporate client of Bank JKY suffers a major financial loss from a cyber-attack and is therefore unable to make payments on its debt to Bank JKY, which triggers a default.

Q-17. A fast-growing UK-based FinTech firm offers savings accounts, cryptocurrency accounts, domestic bill payment services, and services that allow customers to make payments in different currencies. A senior operations manager at the firm is developing a plan to comply with new UK regulatory requirements for operational resilience. Which of the following steps should the manager recommend that the firm take to best comply with the regulatory expectations in this area?

- A. Develop an impact tolerance for each of the internal processes performed by the firm.
- B. Identify important business services and map the dependencies between these services.
- C. Calculate the 1-year 99.9% VaR for operational risk for each of the firm's business divisions and use this result to reserve additional capital for each division.
- D. Create an operational resilience team that is led by the IT department, with its other members coming from the operational risk management, legal, and compliance functions.

Q-18. An external auditor is reviewing the modeling processes used by a US-based bank to model operational losses as part of the bank's capital planning process. Using guidelines set by the Federal Reserve with respect to capital planning, which of the following processes or assumptions would the auditor find most appropriate?

- A. Assuming a high positive correlation between operational loss severity and equity index movements during normal market conditions
- B. Using a net charge-off model to predict shorter-term credit losses and a roll-rate model to predict losses over a longer time horizon
- C. Modeling operational losses by projecting an annual loss estimate and then evenly distributing the losses across the four quarters of the year
- D. Incorporating forward-looking factors and idiosyncratic risk exposures into stressed operational loss estimates

Q-19. An operational risk manager at a large retail bank is asked to review the framework for the bank's risk mitigation controls. As part of this review, the manager classifies the risk controls as preventive, detective, corrective, or directive. Which of the following should the manager classify as a directive control?

- A. An employee training program that explains the policies and procedures for reviewing new account applications
- B. A notification to a credit card customer about a potentially fraudulent transaction on that customer's account
- C. An implementation of an antivirus software update across all of the bank's IT systems
- D. A dual-factor authentication protocol that is used to control access to critical business systems

Q-20. In the context of organizational risk management, controls are categorized based on their purpose and timing relative to potential incidents. Preventive, Detective, Corrective, and Directive Controls serve distinct roles in mitigating risks. Scenario: A financial institution is enhancing its cybersecurity framework. The Chief Information Security Officer (CISO) proposes the following measures:

- Implementing multi-factor authentication (MFA) for system access.
- Deploying intrusion detection systems (IDS) to monitor network traffic.
- Establishing an incident response team to restore systems after a breach.
- Publishing a mandatory cybersecurity policy requiring annual employee training.

Which combination of controls best represents Preventive, Detective, Corrective, and Directive Controls, respectively?

- A. Preventive: MFA; Detective: IDS; Corrective: Incident response team; Directive: None.
- B. Preventive: MFA; Detective: IDS; Corrective: Incident response team; Directive: Cybersecurity policy.
- C. Preventive: Cybersecurity policy; Detective: IDS; Corrective: Incident response team; Directive: MFA.
- D. Preventive: Incident response team; Detective: IDS; Corrective: MFA; Directive:

Cybersecurity policy.

Q-21. An enterprise risk consultant is presenting about the management of risk associated with third-party vendor relationships at a financial conference. To emphasize the importance of understanding this risk and to illustrate lessons learned, the consultant describes several past examples of large losses and data breaches incurred by different financial institutions due to deficient or fraudulent third-party vendor practices. One example provided is the large loss incurred by Capital One, a US-based bank holding company, that resulted from its relationship with a third-party vendor. Which of the following best describes the circumstances that led to the loss in this case?

- A. The vendor provided an inaccurate loan pricing model to Capital One, which incurred far greater default losses than expected.
- B. A bill payment system provided by the vendor failed for an extended period of time, resulting in many Capital One customers canceling their accounts and causing severe reputational impact.
- C. The vendor's sales manager established extremely high incentives for its representatives to sell Capital One products, resulting in regulatory fines for selling inappropriate products to consumers.
- D. A former staff member of the vendor hacked into a database of Capital One's personal customer information that was stored on the vendor's cloud services platform and stole much of this information.

Q-22. The CRO of a regional mortgage lender has asked an enterprise risk manager to develop a set of policies and procedures for the firm's operational risk reporting. The manager considers appropriate policies for the governance of the firm's risk reporting framework and also assesses how the firm should structure its risk reports for different stakeholder groups and organizational functions. Which of the following would be most appropriate for the manager to recommend?

- A. The firm should report a more detailed and extensive set of key risk indicators to the

board of directors than it does to business line managers to support the board's strategic risk review.

- B. The operational risk committee should be responsible for executing all necessary changes to the firm's risk exposures after risk reports are reviewed.
- C. The central operational risk function should be responsible for aggregating information from each of the business units about operational risk exposures.
- D. The firm should emphasize forward-looking risk indicators and avoid the use of backward-looking indicators in its risk reporting.

Q-23. According to bank operational loss data, which of the following Basel II event risk categories most likely results in the greatest loss severity for a financial institution during 2024-2019?

- A. External fraud (EF).
- B. Client, products, and business practices (CPBP).
- C. Employment practices and workplace safety (EPWS).
- D. Execution, delivery, and process management (EDPM).

Q-24. The LDA process is applied to each cluster of losses through a unit of measure (UoM) that needs to be further aggregated into a total loss distribution. Which of the following statements about Units of Measure (UoM) in the context of operational risk is incorrect?

- A. The choice of UoM should reflect how the business is organized and the intended use of the model.
- B. Copulas are used to aggregate UoMs by modeling dependency structures, including correlations and tail dependencies.
- C. To accurately model the tail of operational loss distributions, it is only necessary to complete the dataset with higher value losses.
- D. A trade-off exists between the homogeneity of the data sets and data availability, impacting the precision of the model and its complexity.

3.2. Focus Areas

3.2.1. 重要知识点

3.2.1.1. Types of Model Risk – Model Error

- Assume the distribution of the underlying asset is stationary (volatility).
- Underestimate the number of risk factors
- Perfect capital market assumption.
- Liquidity or rather the absence of liquidity
- Models be misapplied to a given situation.

3.2.1.2. Types of Model Risk – Implementing a Model Wrongly

- Inaccurate data
- Inappropriate length of sampling period
- Problems with liquidity and the bid/ask spread

3.2.1.3. Risks Arising through Outsourcing Activities

- Compliance risks
- Concentration risks
- Reputational risks
- Country risks
- Operational risks
- Legal risks

3.2.1.4. Core Elements of Effective Programs regarding TPRM

- Risk assessments
- Due diligence and selection of service providers
- Contract provisions and considerations
- Incentive compensation review
- Oversight and monitoring of service providers
- Business continuity and contingency plans

3.2.1.5. Three Lines of Defense in ML/FT

- The first line of defence: all banks should implement ongoing employee training programmes so that bank staff are adequately trained to implement the bank's AML/CFT policies and procedures.
- The second line of defence: the chief officer in charge of AML/CFT should have the responsibility for ongoing monitoring of the fulfillment of all AML/CFT duties

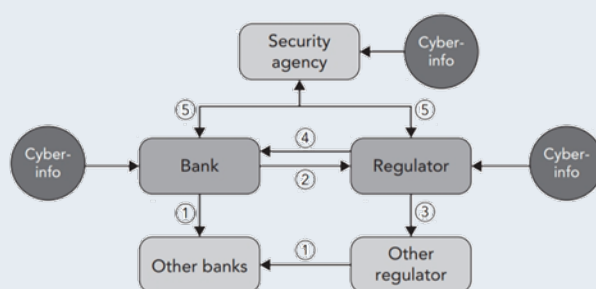
by the bank.

- The third line of defence: internal audit plays an important role in independently evaluating the risk management and controls.

3.2.1.6. Cyber Resilience

- The aim of cyber resilience is to maintain a system's capability to deliver the intended outcome at all times, including times of crisis when regular delivery has failed.

3.2.1.7. Interlinkage of different types of cyber-security information-sharing practices



3.2.2. 基础题

Q-25. You are the head of the Independent Risk Oversight (IRO) unit of XYZ bank, Your first task is to review the following existing policies relating to model implementation.

- I. The remuneration of the staff of the IRO unit is dependent on how frequently the traders of XYZ bank use models vetted by the IRO.
- II. Model specifications assume that markets are perfectly liquid.

Which of the existing policies are sources of model risk?

- A. Statement I only
- B. Statement II only
- C. Both statements are correct
- D. Both statements are incorrect

Q-26. Which of the following about model risk is correct?

- A. When the developer is building the model, all the steps and assumptions should be documented.
- B. There is no need for sensitivity analysis for robustness and stability should likewise be repeated periodically.

- C. With the appropriate procedures and tools, model risk can be eliminated.
 - D. Both the developer and user need to make validation for the model periodically.
- Q-27.** Which of the following scenarios is the best example of a model error?
- A. Assuming a non-normal distribution of returns.
 - B. Assuming perfectly liquid markets.
 - C. Assuming variable distribution of asset price.
 - D. Assuming imperfect capital markets.
- Q-28.** ABC Bank operates in the US and has a service contract in place with XYZ (Service), which operates in France. Service manages a significant amount of confidential customer data for Bank, and recently a computer glitch at Service resulted in the accidental public disclosure of confidential customer data. As a result of the data breach, which of the following risks is Bank least likely to face?
- A. Legal risk.
 - B. Compliance risk.
 - C. Operational risk.
 - D. Country risk.
- Q-29.** The senior management team of a small regional bank has established a committee to review procedures and implement best practices related to entering into significant contracts with third-party vendors. The committee is reviewing one proposed relationship with a third-party vendor who would have a significant responsibility for marketing the bank's financial products to potential customers. In establishing policies to reduce the operational risk associated with this potential vendor contract, which of the following recommendations would be most appropriate?
- A. The bank should review all third-party audit reports of a vendor that are publicly available.
 - B. The bank should ensure that a vendor's sales representatives are compensated mainly with commissions from the sale of the bank's products.
 - C. The bank should prevent a third-party vendor from having access to any of its critical

processes.

- D. The bank should be responsible for developing a vendor's contingency planning process in order to mitigate risk exposure to the vendor.

Q-30. A regional bank is formalizing its policies and procedures to help identify and analyze the risk of potential money laundering transactions. Of special interest is accounting for customers' backgrounds when determining customer acceptance policies. According to Basel Committee guidelines, which of the following correctly describes a best practice that the bank should use in identifying, verifying and profiling customers to help mitigate money laundering risk?

- A. The bank does not need to apply due diligence on a customer if the bank receives funds from that customer's account at another bank that is subject to the same customer due diligence standards.
- B. The bank should apply the same due diligence measures to all customers regardless of their jurisdiction and the nature of their relationship with the bank to prevent discrimination.
- C. The bank should apply its due diligence process not just to potential customers but also to beneficial owners of the proposed customer accounts and persons acting on their behalf.
- D. The bank should not open an account for or conduct business with a customer who wants to remain anonymous to the bank except for confidential "numbered accounts" that function as anonymous.

Q-31. A person named Jack Ryan has applied to open a new account at Quadstreet International Bank. The bank has a modern, well-established, and compliant customer acceptance policy. Immediately the bank is able to determine two facts: Jack is a high-risk customer, but he previously had an account at another large bank. Further, Quadstreet does conduct business with Jack's previous bank and considers it to be reputable. With respect to money laundering and terrorism financing (ML/FT), which of

the following statements is TRUE about Jack Ryan's application ?

- A. Jack's identity must be established and verified before carrying out any transactions
- B. Because Jack is a high-risk customer, the bank is required to avoid him like any high-risk customer
- C. Because Jack already has an account at a reputable bank, he can be classified as low risk and well-identified
- D. If Jack is a politically exposed person (PEP), he can be classified as low-risk on the assumption that the media already vets him and he cannot evade scrutiny.

Q-32. Which of the following statements regarding the role of employees in making an organization more cyber resilient is correct ?

- A. Effective cyber training requires the full buy-in of all employees.
- B. Effective gamification requires mandatory participation of the employees.
- C. To date, studies of incentive training have shown incentivized training yields limited but positive results.
- D. Treating cyber security as a competitive game would not be considered appropriate.

Q-33. Which cyber resiliency control is least prone to false positives or false negatives?

- A. Penetration testing.
- B. Anomaly detection algorithm.
- C. Signature-based detection method.
- D. Network behavior anomaly detection.

Q-34. Which of the following statements is TRUE in regard to impact tolerance?

- A. Impact tolerance refers to the risk of loss from inadequate or failed processes, people or systems or from external events.
- B. The key drawback of impact tolerances is their irrelevance to outsourced (third-party) service providers who cannot be directly controlled.
- C. Impact tolerances describe the maximum tolerable level of disruption to an important

business service or an important group business service, as measured by a length of time in addition to any other relevant metrics.

- D. Because the set of impact tolerances is identical to the firm's risk appetite, for regulatory purposes the firm's risk appetite statement is an acceptable substitute for its impact tolerance statement.

Q-35. A regional bank follows the three lines of defense approach for managing its operational risk. The CRO of the bank is concerned that several competing banks have recently received regulatory fines resulting from poor management of risks related to money laundering and financing of terrorism (ML/FT), and wants to improve the bank's management of these risks. The CRO reviews the Basel Committee guidelines for the management of ML/FT risks and suggests that the bank adopt industry best practices for different business functions in each line of defense. Which of the following would be most appropriate for the CRO to recommend addressing ML/FT risks?

- A. The chief officer in charge of ML/FT risk should be responsible for monitoring and managing these risks within each of the business lines on a daily basis.
- B. The CRO should serve as the primary point of contact with regulatory supervisors on any issues related to ML/FT risk.
- C. Internal and external auditors should evaluate the bank's risk controls to ensure that they are effective in managing ML/FT risk as well as compliant with regulations.
- D. The CORF should screen current and prospective employees as part of the second line of defense in managing ML/FT risk.

Q-36. A risk consultant is meeting with the senior management of a midsize bank to discuss ways in which the bank can improve its management of financial crime and fraud risk. During the meeting, the consultant uses the example of USAA Bank to explain how weaknesses in controls and processes used to manage these risks can result in large losses as well as regulatory fines. Which of the following best explains the fine imposed by regulators on USAA Bank?

- A. USAA Bank was fined after many retail bankers were found to have opened unwanted accounts for existing customers in order to meet performance incentives.
- B. USAA Bank was fined for failing to correct deficiencies identified by regulators in its management and reporting of suspicious transactions after an extended period of time.
- C. USAA Bank was fined after several subsidiaries were found to have laundered over USD 1 billion in funds belonging to large criminal organizations.
- D. USAA Bank was fined after fraudulently marketing complex interest rate derivatives to large municipal clients.

Q-37. The board of directors of a fast-growing US-based bank is discussing a report of findings and recommendations issued after a recent regulatory review of the bank's risk management framework. A board member notes that the regulators have identified deficiencies in the bank's model validation process, and recommends that the CRO develop a plan for improvement in this area. The CRO consults the Federal Reserve guidelines for examples of best practices, and identifies ways that the bank can improve its validation approaches to conform with the guidelines. Which of the following recommendations is appropriate for the CRO to make?

- A. Require internal auditors that are responsible for assessing the bank's model risk management framework to also validate and backtest the models.
- B. Begin monitoring the performance of each model only after the backtest of the model has been successfully completed.
- C. Ensure that third-party vendor models used by the bank are implemented under the assumptions that were initially provided by the vendor.
- D. Establish a continuous process of model risk management and validation rather than a process that conducts reviews at specific points in time.

Q-38. A midsize bank specializing in residential mortgages and credit cards is planning to offer a new commercial loan product. The bank has an existing credit rating model for its residential mortgage products which has performed well and has been successfully back-

tested over several years, but the CRO considers whether the bank should develop and implement a new model for the commercial loan product. In addition, the CRO wants to ensure that the bank follows best practices for the development, validation, and implementation of any models that it uses. Which of the following actions should the CRO recommend that the bank take?

- A. Apply the strongly performing existing model to the new commercial loan product given the model's successful track record.
- B. Develop a new model for the commercial loan product and avoid the use of qualitative or judgmental adjustments to the model's quantitative output.
- C. Make the model development team responsible for validating all of the bank's models that are currently operational.
- D. Perform sensitivity analysis on all models used by the bank to identify market conditions under which they might perform poorly.

Q-39. The CRO of a regional bank wants to ensure that the modeling assumptions used in the bank's economic capital models are sound. The CRO asks a member of the validation team to review the bank's process of assessing the interest rate risk in its banking book and to validate the assumptions used in its interest rate models. Which of the following assumptions would be most appropriate for the bank to make?

- A. The bank changes the interest rate it offers to depositors by the full amount of any change in market interest rates.
- B. The bank models its retail non-maturity deposits as floating-rate, putable bonds.
- C. The bank assumes that its residential mortgages exhibit positive convexity as interest rates decrease.
- D. The bank models its interest rate risk in the banking book independently from its credit risk.

Q-40. A bank wants to reduce its operating expenses and considers hiring a third-party service provider to offer additional loan origination and credit services to some of the bank's

customers. The bank's legal department has begun negotiating the terms of a contract with the provider. Which of the following describes the most appropriate set of due diligence actions for the bank to take before signing the contract?

- A. The bank should audit the service provider's operational processes and also give the service provider the similar right to audit the bank's processes.
- B. The bank should purchase insurance to cover potential losses resulting from the provider's services and should require that the provider deposit collateral with the bank to mitigate performance risk.
- C. The bank should determine the compensation structure for the provider's sales representatives and ensure that it incentivizes their productivity through a high proportion of variable compensation.
- D. The bank should define specific events that are considered a default under the contract and give the provider an opportunity to resolve a default before terminating the contract.

Q-41. The CRO at a bank wants to strengthen the bank's capability to defend itself against emerging cyber-threats. To help achieve this goal, the CRO is assessing the current range of practices regarding the sharing of cybersecurity information between different types of institutions, as well as the potential benefits from sharing information. Which of the following statements would be most appropriate for the CRO to make?

- A. The sharing of cybersecurity information among banks is less frequently observed and generally considered to be less effective than other cyber-security information-sharing practices.
- B. The scope and depth of information-sharing practices among banks may significantly vary between financial markets, depending on the level of trust among participating banks.
- C. Information-sharing among different national regulators has evolved significantly over the past several years and is now a widespread practice at a large majority of jurisdictions.
- D. Existing peer-sharing mechanisms among banks focus on the exchange of information related to cyber-security incidents, but such information is generally not shared from

banks to regulators.

Q-42. The board of directors at a large bank wants to improve the bank's practices for managing money laundering and financial terrorism (ML/FT) risk. The risk committee of the bank meets to discuss ways to achieve this objective that conform to best practices. Which of the following actions would be most appropriate for the bank to recommend?

- A. Require the bank's business units to screen potential employees as part of the first line of defense in managing ML/FT risk.
- B. Establish a threshold transaction value and review all transactions above this threshold for evidence of ML/FT.
- C. Exclude politically exposed persons (PEPs) from screening for ML/FT risk due to their much lower ML/FT risk.
- D. Give the compliance and legal functions the primary responsibility for managing ML/FT risk.

Q-43. From a consumer protection perspective, which of the following statements is incorrect?

- A. MiFID I introduced pre-trade transparency requirements to improve market efficiency and protect investors.
- B. MiFID II expanded on the consumer protection measures of MiFID I by including rules on product governance and independent investment advice.
- C. The Dodd-Frank Act was primarily focused on reducing systemic risk and did not include provisions related to consumer protection.
- D. MiFID II requires firms to ensure that financial products are marketed and sold only to clients for whom they are deemed suitable.

Q-44.  A multinational corporation is evaluating cybersecurity frameworks to enhance its risk management strategy. The Chief Information Security Officer (CISO) outlines the following requirements:

1. A voluntary framework that provides best practices for identifying, protecting,

detecting, responding to, and recovering from cyber threats.

2. A prioritized list of actionable controls to mitigate prevalent attacks, regularly updated for modern threats like cloud computing and remote work.
3. An internationally recognized certification standard requiring systematic risk management through an Information Security Management System (ISMS) and the PDCA cycle.

Which combination of cybersecurity standards best aligns with these requirements?

- A. 1. NIST CSF; 2. ISO/IEC 27001; 3. CIS Controls
- B. 1. CIS Controls; 2. NIST CSF; 3. ISO/IEC 27001
- C. 1. NIST CSF; 2. CIS Controls; 3. ISO/IEC 27001
- D. 1. ISO/IEC 27001; 2. CIS Controls; 3. NIST CSF

3.3. Capital Planning

3.3.1. 重要知识点

3.3.1.1. The RAROC measure is essential to successful integrated risk management. Its main function is to relate the return on capital to the riskiness of firm investments. The RAROC is the risk-adjusted return divided by risk-adjusted capital (e.g., economic capital).

3.3.1.2. $\text{RAROC} = (\text{expected revenues} - \text{costs} - \text{expected losses} - \text{taxes} + \text{return on economic capital} \pm \text{transfer price}) / \text{economic capital}$

3.3.1.3. The hurdle rate is computed as follows:

$$h_{AT} = \frac{(CE \times R_{CE}) + (PE \times R_{PE})}{CE + PE}$$

- If $\text{RAROC} > \text{hurdle rate}$, there is value creation from the project and it should be accepted.
- If $\text{RAROC} < \text{hurdle rate}$, there is value destruction from the project and it should be rejected.
- An adjusted RAROC (ARAROC) measure was developed to better align the risk of the business with the risk of the firm's equity.

3.3.1.4. $\text{Adjusted RAROC} = \text{RAROC} - \beta_E(R_M - R_F)$

- If adjusted RAROC $> R_F$, then accept the project
- If adjusted RAROC $< R_F$, then reject the project

3.3.1.5. Problem of coherence when designing scenarios: problems are inherently multi-factored, making it more difficult to design a coherent stress test. It is not sufficient to specify one potential problem because the others do not remain fixed.

3.3.2. 基础题

Q-45. A risk manager is considering a HKD 400 million loan that will be fully funded by deposits paying an average annual interest rate of 1.0%. The risk manager has estimated the following regarding the loan:

Expected annual revenue:	HKD 16 million
Expected loss:	HKD 1.0 million
Unexpected loss:	HKD 48.0 million
Economic capital required:	HKD 47.0 million
Annual operating expenses:	HKD 2.0 million

Assuming the economic capital can be invested so that it earns 2% per year, the risk manager correctly calculates the risk-adjusted return on capital (RAROC) to be 21.1%. Assuming nothing else changes, which of the following would increase the RAROC estimate the most?

- A. Annual operating expenses decrease by 50%.
- B. Expected annual revenue increases by 20%.
- C. The economic capital can be invested so that it earns 4% per year.
- D. The loan is fully funded by deposits paying an average annual interest rate of 0.5%.

Q-46. Which of the following statements regarding the risk-adjusted return on capital (RAROC) methodology is correct?

- A. In the context of performance measurement, RAROC uses accounting profits.
- B. In the numerator of the RAROC equation, expected loss is added to the return.
- C. If a business unit's cost of equity is greater than its RAROC, then the business unit is not adding value to shareholders.

- D. RAROC is useful for determining incentive compensation but it lacks the flexibility to consider deferred or contingent compensation.

Q-47. Assume a bank's \$2.0 billion corporate loan portfolio offers a return of 6% per annum. The expected loss on the portfolio is estimated to be 1.5% per annum; i.e., \$30 million. The portfolio is funded by \$2 billion in retail deposits with a transfer-priced interest rate charge of 2%. The bank has a direct operating cost of \$16 million per annum and an effective tax rate of 25%. Risk analysis of unexpected losses associated with the portfolio tell us we need to set aside economic capital of \$200 million against the portfolio; i.e., 10% of the loan amount. The bank's economic capital must be invested in risk-free securities and, unfortunately in the regime of ultra low interest rates, the risk-free rate on government securities is only 1%. Although the loan portfolio's risk-adjusted return on capital (RAROC) is positive and seemingly high, the bank wants to adjust the traditional RAROC calculation to obtain a RAROC measure that takes into account the systemic riskiness of the expected returns. If the risk-free rate is 1%, and the expected rate of return on the market portfolio is 8% such that the equity risk premium is 7%, and the beta of the firm's equity is 1.6, which of the following is the correct adjusted RAROC and is the project advisable?

- A. RAROC is 6.25% but no, the project is bad because RAROC is below the risk-free rate.
B. RAROC is 8% but no, the project is bad because RAROC is below the risk-free rate.
C. RAROC is 9.8% and yes, the project is good because RAROC is above the risk-free rate.
D. RAROC is 13.5% and yes, the project is good because RAROC is above the risk-free rate.

Q-48. ABC Corp. is attempting to compute a single hurdle rate to use for all of its business activities. The following information has been provided:

Market value of common equity	\$500 million
Market value of preferred equity	\$300 million
Beta of common equity	1.2

After-tax RAROC of proposed project	12%
Risk-free rate	2%
Cost of preferred equity	4%
Expected return on market portfolio	7%

Based on the information provided, which of the following statements is correct?

- A. ABC's hurdle rate is 6.5%
- B. ABC's hurdle rate is 6.2%
- C. ABC's hurdle rate is 5.5%
- D. ABC's hurdle rate is 5.8%

Q-49. A packaging materials manufacturer is considering a project that has an estimated risk-adjusted return on capital (RAROC) of 15%. Suppose that the risk-free interest rate is 3% per year, the expected market rate of return is 11% per year, and the company's equity beta is 1.8. The manufacturer uses the adjusted RAROC metric as the criterion to decide whether or not to accept the project. Which of the following correctly describes the decision the company should make and the rationale for making that decision?

- A. Reject the project because the adjusted RAROC is higher than the market expected excess return.
- B. Accept the project because the adjusted RAROC is higher than the market expected excess return.
- C. Reject the project because the adjusted RAROC is lower than the risk-free interest rate.
- D. Accept the project because the adjusted RAROC is lower than the risk-free interest rate.

Q-50. A regional commercial bank is considering a 1-year loan to be fully funded by deposits, with the following parameters:

- Loan amount: JPY 4.2 billion
- Average annual interest rate paid on deposits: 0.4%
- Annual interest rate received on loan: 3.2%

- Expected loss: 2.0% of face value of loan
- Annual operating costs: 0.5% of face value of loan
- Economic capital required to support the loan: 10.0%
- Average pre-tax return on economic capital: 1.4%
- Effective tax rate: 38%
- Other transfer costs: JPY 0

What is the after-tax RAROC for this loan?

- A. 0.27%
- B. 2.73%
- C. 4.40%
- D. 10.73%

Q-51. Piper Hook, a bank examiner, is trying to make sense of stress tests done by one of the banks she examines. The stress tests are multi-factored and complex. The bank is using multiple extreme scenarios to test capital adequacy, making it difficult for Hook to interpret the results. One of the key stress test design challenges that Hook must deal with in her examination of stress tests is:

- A. Multiplicity
- B. Efficiency
- C. Coherence
- D. Efficacy

Q-52. Nordlandia is a country with a developed economy maintaining its own currency, the Nordlandian crown (NLC), and whose most important export is domestically produced oil and natural gas. In a recent stress test of Nordlandia's banking system, several scenarios were considered. Which of the following is most consistent with being part of a coherent scenario?

- A. An increase in domestic inflation and appreciation of the NLC.
- B. A significant increase in crude oil prices and a decrease in the Nordlandian housing price

index.

- C. A drop in crude oil prices and appreciation of the NLC.
- D. A sustained decrease in natural gas prices and a decrease in the Nordlandian stock index.

Q-53. A risk consultant is presenting on the evolution of macro-prudential stress testing requirements established in response to the global financial crisis of 2007 – 2009. The consultant compares features of the following three stress tests:

- The 2009 Supervisory Capital Assessment Program (SCAP) test
- The 2011 European Banking Association (EBA) test
- The 2012 Comprehensive Capital Analysis and Review (CCAR) test

Which of the following elements of these stress tests or their resulting impacts is correct for the analyst to include in the presentation?

- A. Several Spanish banks failed the EBA stress test, and Spain imposed stricter countrywide stress tests the following year that resulted in some banks raising capital.
- B. The SCAP stress test scenario was less severe compared to later US stress tests and did not result in any increased capital requirements for participating banks.
- C. The CCAR test required banks to evaluate both a base case and a stress scenario, while the EBA test only included a stress scenario.
- D. The SCAP test only disclosed overall loss rates for retail and commercial exposures, while the EBA test expanded disclosure to individual geographical regions and asset classes.

3.4. Basel Accords

3.4.1. 重要知识点

3.4.1.1. Evolution of Basel Accord

- The Basel Capital Accord (Basel I), concluded on July 15, 1988, set capital charges against credit risk based on a set of relatively simple rules.
- Capital adequacy requirements against market risk were added to Basel I in 1996.
- In June 2004, Basel II created more risk-sensitive capital requirements and added a charge against operational risks.

- During the credit crisis, it was recognized that some changes were necessary to the calculation of capital for market risk in the Basel II framework. These changes are referred to as “Basel 2.5.”
- The credit crisis that started in 2007 revealed serious weaknesses in the regulatory framework. This led to a new round of revisions, which are informally called Basel III.

3.4.1.2. Pillar 1: Minimum Capital Requirements

- **Capital Ratio:**

$$\frac{\text{Total Capital}}{\text{Credit Risk} + \text{Market Risk} + \text{Operational Risk}} = \text{Bank's Capital Ratio} \geq 8\%$$

$$\frac{\text{Total Capital}}{\text{RWA}_{\text{Credit}} + (\text{MRC}_{\text{Market}} \times 12.5) + (\text{ORC}_{\text{Operational}} \times 12.5)} \geq 8\%$$

- **Capital Requirements:**

Basel III increases the minimum capital ratio for core tier 1 capital from 2% to 4.5%. Tier 1 capital is increased from 4% to 6%. Total capital is still kept at a minimum of 8%.

- **Common Equity Tier 1 Capital:**

- Common shares issued by the bank.
- Stock surplus (paid-in capital)
- Retained earnings.
- Minority interests.
- Accumulated and other comprehensive income and other disclosed reserves.
- Regulatory adjustments applied to the calculation of Common Equity Tier 1 capital.

3.4.1.3. Pillar 2: Supervisory Review Process

3.4.1.4. Pillar 3: Market Discipline

3.4.1.5. The standardized approach for Credit Risk incorporates risk weights based on external credit rating assessments. The amount of capital that a bank must hold is specific to the risk of credit-risky assets, the type of institution the claim is written on, and the maturity of those assets.

3.4.1.6. Standardized method for Market Risk:

Determines capital charges associated with various market risk exposures (equity risk, interest rate risk, foreign exchange risk, commodity risk, and option risk). The market risk capital charge for each market risk is computed as 8% of its market-risky assets.

3.4.1.7. Internal models approach (IMA) for Market Risk:

Allows a bank to use its own risk management systems to determine its market risk capital charge. The market risk charge is the higher of (1) the previous day's VaR or (2) the average VaR over the last 60 business days adjusted by a multiplicative factor (subject to a floor of 3).

$$3.4.1.8. \text{MRC}_t^{\text{IMA}} = \text{Max} \left(k \frac{1}{60} \sum_{i=1}^{60} \text{VAR}_{t-i}, \text{VAR}_{t-1} \right) + \text{Max} \left(k_s \frac{1}{60} \sum_{i=1}^{60} \text{SVAR}_{t-i}, \text{SVAR}_{t-1} \right) + \text{SRC}_t + \text{IRC}_t$$

- SVaR is the Stress VaR.
- SRC is the specific risk charge, which is a buffer against idiosyncratic factors, including basis risk and event risk.
- IRC is an incremental risk charge that covers: (1) default risk, (2) credit migration risk for debt instruments. IRC is calibrated to a 99.9% confidence level over one year.

3.4.1.9. An exception occurs if the day's change in value exceeded the VaR estimate of the previous day. When backtesting VaR, the number of exceptions is determined for a 250-day testing period. Based on the number of exceptions, the bank's exposure is categorized into one of three zones and VaR is scaled up by the appropriate multiplier.

- Green zone: 0-4 exceptions, increase in exposure multiplier is 0.
- Yellow zone: 5-9 exceptions, exposure multiplier increases between 0.4 and 0.85.
- Red zone: Greater than or equal to 10 exceptions, multiplier increases by 1.

3.4.1.10. SCR and MCR of Solvency II.

- If its capital falls below the SCR level, an insurance company should, at minimum, deliver to the supervisor a plan to restore capital to above the SCR level. The supervisor might require the insurance company to take particular measures to correct the situation.
- The MCR is regarded as an absolute minimum level of capital. If capital drops below the MCR level, supervisors may prevent the insurance company from taking new business.

It might force the insurance company into liquidation, transferring its policies to another company. The MCR will typically be between 25% and 45% of the SCR.

- Two ways to calculate the SCR: the standardized approach and the internal models approach.

3.4.1.11. Capital Conservation Buffer will be required to provide an extra cushion against loss in times of stress. The buffer will be an additional 2.5% Common Equity Tier 1 capital requirement.

3.4.1.12. Banks will be subject to a **countercyclical buffer** if regulatory authorities deem it necessary. The buffer is intended to protect the banking sector by ensuring that capital requirements take into account macro-environment factors.

3.4.1.13. The committee has introduced a non-risk based **leverage ratio** that will act as a supplementary measure to risk-based capital standards. The goals of the leverage ratio are to constrain the build-up of leverage in the banking sector and to provide a simple “back-stop” measure of leverage that supplements and reinforces risk-based capital standards. The leverage ratio of 3% (Tier 1 capital to on- and off-balance sheet items and exposures) is targeted to take effect January 1, 2018.

3.4.1.14. Goal: ensure banks have adequate, high-quality liquid assets to survive short-term stress scenario.

$$LCR = \frac{\text{high - quality liquid assets}}{\text{total net cash outflows over the next 30 calendar days}} \geq 100\%$$

3.4.1.15. Goal: protect banks over a longer time horizon than LCR.

$$NSFR = \frac{\text{Amount of Stable Funding}}{\text{Required Amount of Stable Funding}} \geq 100\%$$

3.4.1.16. Contingent Convertible Bonds (CoCos)

- Contingent convertibles, also known as CoCo bonds, CoCos or contingent convertible notes, are slightly different to regular convertible bonds because bonds converting to equity are "contingent" on a pre-specified event, such as falling down of bank's Tier 1 capital below a certain percentage vis-à-vis its risk-weighted assets.

3.4.1.17. Basel III Reforms

- The revisions seek to restore credibility in the calculation of risk-weighted assets and improve the comparability of banks' capital ratios by:
- Enhance the standardised approaches for credit risk, credit valuation adjustment (CVA) risk and operational risk.
- Constrain the use of the internal model approaches.
- Introduce a leverage ratio buffer to further limit the leverage of global systemically important banks (G-SIBs).
- Place a more robust risk-sensitive floor.

3.4.1.18. SMA Capital Requirement

- The SMA combines the Business Indicator (BI) Component and Internal Loss Component.
- BI is a financial statement proxy of operational risk exposure. To compute the BI for year t, bank must determine the 3 year average of the BI, as the sum of the 3 year average of its components.
- $BI = ILDC_{Avg} + SC_{Avg} + FC_{Avg}$
- Banks are divided into three buckets according to the size of their BI.

Bucket	BI Range in Euro(bn)	BI Marginal Coefficients
1	≤ 1	12%
2	$1 < BI \leq 30$	15%
3	> 30	18%

- Internal Loss Component: The Loss Component (LC) = 15 * average annual operational risk losses incurred over the previous 10 years.
- $ILM = \ln[\exp(1) - 1 + (\frac{LC}{BIC})^{0.8}]$
- The SA OP Risk Capital Requirement:

$$ORC = BIC \times ILM$$

3.4.2. 基础题

Q-54. A newly established risk division of a regional financial institution is setting up a Monte

Carlo simulation methodology to estimate the firm's aggregate loss distribution. Which of the following loss frequency and loss severity distribution pairs is the most appropriate to use?

- A. Binomial distribution for frequency, and Poisson distribution for severity.
- B. Lognormal distribution for frequency, and Weibull distribution for severity.
- C. Negative Binomial distribution for frequency, and Pareto distribution for severity.
- D. Transformed Beta distribution for frequency, and Normal distribution for severity.

Q-55. Your bank is implementing the advanced Internal Rating Based Approach of Basel II for credit risk, and the Advanced Measurement Approach for operational risk. The bank uses the Internal Model Approach for market risk. The Chief Risk Officer (CRO) wants to estimate the bank's total risk by adding up the regulatory capital for market risk, credit risk, and operational risk. The CRO asks you to identify the problems with using this approach to estimate the bank's total risk. Which of the following statements about this approach is incorrect?

- A. It assumes market, credit, and operational risks have zero correlation.
- B. It uses a 10-day horizon for market risk.
- C. It ignores strategic risks.
- D. It ignores the interest risk associated with the bank's loans.

Q-56. In regard to Basel II minimum capital requirements, which of the following is false?

- A. Banks can reduce their capital charge, subject to a limit, if they can demonstrate diversification benefit due to imperfect correlation between the major risk buckets: credit, operational and market risk.
- B. Pillar Two explicitly encourages national authorities (supervisors) to supplement Pillar One with additional capital requirements at their discretion if they deem appropriate
- C. Under the advanced/internal approaches, all three risk categories (credit, market, and operational risk) employ value at risk (VaR) concepts
- D. Basel II had no explicit charge for liquidity risk

- Q-57.** Each of the following is a required disclosure element (i.e., must be disclosed by the bank) of the Third Pillar (Pillar 3) except for:
- A. The bank's capital structure including a break-down of Tier 1 capital, deductions, and total eligible capital
 - B. Details of the bank's compensation (remuneration) program including performance metrics, risk metrics, and variable pay plan elements
 - C. The bank must disclose securitization special purpose entities (SPEs) even if the bank is only a sponsor (i.e., only "manages or advises" on the placement of securities)
 - D. The bank's long-term strategy and a mapping of new product development initiatives to planned target markets and customers
- Q-58.** Pillar 1 of the Basel II framework allows banks to use various approaches to calculate the capital requirements for credit risk, operational risk, and market risk. Which of the following Basel II approaches allows a bank to use its own estimates of recovery rates?
- A. The standardized measurement approach for operational risk
 - B. The advanced internal ratings-based approach for credit risk
 - C. The foundation internal ratings-based approach for credit risk
 - D. The fundamental review of the trading book (FRTB) approach for securitized products
- Q-59.** John Smith is a bank supervisor responsible for the oversight of Everbright Group, a large banking conglomerate. Everbright Group now determines its credit risk profile according to the foundation IRB approach and assesses operational risk according to the standardized approach as described in the Basel II Capital Accord. Which of the following are specific issues that should be addressed as part of Smith's supervisory review process of Everbright Group?
- I. Review the bank's internal control systems.
 - II. Check compliance with transparency requirements as described in Pillar 3 of Basel II Accord.

- III. Make sure that the bank estimates for LGD and EAD for its corporate loans are in compliance with supervisory estimates.
- IV. Evaluate the impact of interest rate risk by assessing the impact of a 200 basis Point interest rate shock to the bank's capital position.

- A. I and III only
- B. II and IV only
- C. I, II, and IV only
- D. I, II, III, and IV

Q-60. A bank uses the basic indicator approach (BIA) to determine their capital charge for operational risk under Basel II (or Basel III). The bank's annual gross income (GI) over the previous three years was +\$130 million (T-3), -\$60 million loss (T-2), and +\$230 million (T-1). What is the bank's operational risk capital charge?

- A. \$15.0 million
- B. \$18.0 million
- C. \$27.0 million
- D. \$34.5 million

Q-61. A bank uses the standardized approach (SA) to determine their capital charge for operational risk under Basel II (or Basel III). The bank has three (3) business lines and each business line contributes one-third toward the total gross income. For a given total gross income, which business mix will produce the largest capital charge?

- A. Corporate finance; trading and sales; payment and settlement
- B. Retail banking; retail brokerage; and asset management
- C. Commercial banking; agency services; asset management
- D. Retail banking; Commercial banking; and Payment and settlement

Q-62. Each of the following is true about the Basic Indicator Approach (BIA) to operational risk under Basel III except which is false?

- A. Operational risk is the risk of loss resulting from inadequate or failed internal processes, people and systems or from external events
- B. The definition of operational risk includes strategic and reputational risk, but excludes legal risk
- C. Bank's using the Basic Indicator Approach (BIA) must hold capital for operational risk equal to the average over the previous three years of a fixed percentage (denoted alpha) of positive annual gross income
- D. Under the Basic Indicator Approach (BIA), Gross Income is defined as net interest income plus net non-interest income

Q-63. Each of the following is true about the Standardized Approach (SA) to operational risk under Basel III except which is false?

- A. Whereas the Basic Indicator Approach (BIA) use Gross Income for the whole institution as a proxy for the scale of business operations, the standardized approach (SA) calculates the capital charge for each business line by multiplying its gross income by a factor (denoted beta) assigned to that business line
- B. The beta factor in the Standardized Approach (SA) serves as a proxy for the industrywide relationship between the operational risk loss experience for a given business line and the aggregate level of gross income for that business line
- C. Under the standardized approach (SA), business units that fail to provision expected operational losses must calibrate their risk charge based on the unexpected loss at 99.99% confidence level (i.e., rather than 99.9%) over a one-year horizon
- D. A national supervisor can allow a bank to use the Alternative Standardized Approach (ASA) which replaces gross income with loans and advances for retail and commercial banking business lines

Q-64. Each of the following is true about Advanced Measurement Approach (AMA) to operational risk under Basel III except which is false?

- A. Under the AMA, a bank must develop specific policies and have documented criteria for

mapping gross income for current business lines and activities into the AMA framework.

- B. Under the AMA,a bank can use its own internal model(s) but the quantitative standards include a charge for unexpected losses at 99.9% confidence over one-year horizon.
- C. The most popular distribution selection for modeling frequency is the Poisson distribution, the most common and least complex approach to modeling severity is to use a lognormal distribution.
- D. To qualify for the AMA,the bank must : use internal loss data,external loss data, scenario analysis; and take into account key business environment and internal control factors.

Q-65. Pillar 1 of the Basel II framework allows banks to use various approaches to calculate the capital requirements for credit risk, operational risk, and market risk. Which of the following Basel II approaches allows a bank to explicitly recognize diversification benefits?

- A. The basic indicator approach for operational risk
- B. The standardized approach for market risk
- C. The internal models approach for market risk
- D. The standardized approach for operational risk

Q-66. A bank uses VaR and stressed VaR market risk framework in line with the Basel requirements. The bank's internal models for market risk have generated the following risk measures (in USD million) for the current trading book positions:

Confidence Level	Lateset Available 10-day VaR	Latest Available 10-day Stressed VaR	Average 10-day VaR of Previous 60 Days	Average 10-day Stressed VaR of Previous 60 Days
95.0%	238	484	252	546
99.0%	451	995	413	1,106
99.9%	578	1,281	528	1,372

Assuming the supervisory authority has set the multiplication factors for both the VaR and stressed VaR values to 3, what is the correct capital requirement for general market risk for the bank under Basel II.5?

- A. USD 1,248 million
- B. USD 1,533 million
- C. USD 4,557 million
- D. USD 4,799 million

Q-67. The Basel II accord requires a supervisory backtesting framework with all of the following components except:

- A. Seven zones with different plus factors.
- B. Find out the number of exceptions.
- C. Extends over a 1-year period (i.e., 250 trading days).
- D. A multiplier that is subject to a floor of three.

Q-68. Each of the following was both (i) a deficiency or omission of Basel II but is, at the same time, (ii) explicitly addressed by new requirement in Basel III except for

- A. Basel II did not formally include liquidity risk, but Basel III explicitly covers liquidity risk
- B. Basel II could arguably create a procyclical effect, but Basel III explicitly adds a buffer to address this
- C. Basel II did not require external credit ratings, but Basel III seeks to increase the reliance on external ratings
- D. Basel II allowed many banks to show strong risk-based regulatory capital ratios despite high on- and off-balance sheet leverage; Basel III adds a simple leverage ratio to act as a backstop to the risk-based capital ratio

Q-69. Assume a bank determines credit risk-weighted assets (credit RWA) of \$10 million, a market risk charge (MRC) of \$300,000 and an operational risk charge (ORC) of \$500,000. To meet Basel III requirements, the bank has determined it holds \$2.0 million in eligible total (Tier 1 plus Tier 2) regulatory capital. What is the bank's total capital ratio?

- A. 5.0 %
- B. 6.25%

- C. 8.0%
- D. 10.0%

Q-70. With respect to Basel II, Basel III immediately (i.e., effective in 2011 regardless of phase-in arrangements) changes or adds each of the following except for:

- A. Eliminated Tier 3 capital
- B. Restricted the definition of Tier 1 capital
- C. Increased the (Pillar One) Minimum Total Capital (Tier 1 + Tier 2) requirement
- D. Adds a capital conservation buffer (CCB) where none existed in Basel II

Q-71. In updating the Basel II regulatory framework, the Committee asserted that Basel III introduced “a number of fundamental reforms to the international regulatory framework.” Each of the following was a brand new introduction by Basel III (with respect to Basel II) except which was not?

- A. Liquidity ratios were newly introduced in Basel III
- B. A leverage ratio was newly introduced in Basel III
- C. The incremental risk charge (IRC) was newly introduced in Basel III
- D. A credit value adjustment (CVA) charge was newly introduced in Basel III

Q-72. Which statement is true regarding Common Equity Tier 1 capital?

- A. Common Equity Tier 1 capital to risk-weighted assets must be 6% beginning January 1, 2015.
- B. Preferred stock will make up the bulk of Common Equity Tier 1 capital because shareholders cannot force the bank into bankruptcy.
- C. Common Equity Tier 1 capital has the least-stringent requirements for what constitutes capital.
- D. To qualify as common shares that may be used for Common Equity Tier 1 capital, investors of the shares must have a residual claim to the assets.

Q-73. Thrift Bank carries risk-weighted assets (RWA) of \$40.0 billion. In regard to its eligible regulatory capital, the bank holds:

- \$2.8 billion of Common Equity Tier 1 Capital (“Core Tier 1”)
- \$0.2 billion of Additional Tier 1 Capital
- \$1.4 billion of Tier 2 Capital (“Gone concern”)

Does Thrift Bank meet the Basel III capital requirements?

- A. No, because Tier 1 Capital is not at least 8.5%
- B. No, because Total Capital is not at least 10.5%
- C. Yes, because Tier 1 is at least 4.0%
- D. Yes, because Tier 2 is at least 2.5%

Q-74. In the latest guidelines for computing capital for incremental risk in the trading book, the incremental risk charge (IRC) addresses a number of perceived shortcomings in the 99 %/10-day VaR framework. Which of the following statements about the IRC is/are correct?

- I. For all IRC-covered positions, the IRC model must measure losses due to default and migration over a one-year horizon at a 99% confidence level.
 - II. A bank can incorporate into its IRC model any securitization positions that hedge underlying credit instruments held in the trading account.
 - III. The IRC requires banks to calculate a one-year 99.9% VaR for losses from credit-sensitive products in the trading book taking both credit rating changes and defaults into account.
- A. I and II
 - B. III
 - C. I, II, and III
 - D. II and III

Q-75. Which of the following characteristics outlined describe the measurement of stressed value at risk?

- A. The stressed VaR is calculated with one year time horizon.
- B. Historical bank data from the same portfolio is used in measuring SVaR.
- C. The stressed confidence interval is a 95% one-tailed test.
- D. The multiplication factor used in calculating SVaR is the same as that for VaR.

Q-76. Which of the following is true about the standardized measurement method for the calculation of market risk under Basel III?

- A. Tier 3 capital is eligible to support market risks calculated by the standardized approach in Basel III
- B. The capital charge is an arithmetic sum of charges across categories, including interest rate risk, equity position risk, foreign exchange risk, commodities risk, and options risk
- C. Banks have the flexibility to use IMA or SA method to calculate market risk capital.
- D. The standardized measurement method for the calculation of market risk under Basel Accord takes the benefits of diversification into account.

Q-77. Under Basel III, each of the following is true about the internal models approach (IMA) to market risk except which is false?

- A. Value at risk (VaR) must be computed on a daily basis with a one-tailed confidence level of 99.0% and a minimum holding period of ten (10) days
- B. In 2008, the Basel Committee recognized that most of the losses in the credit market turmoil of 2007 and 2008 were from changes in credit ratings, widening of credit spreads, and loss of liquidity, but as a result of defaults excluded.
- C. Regulators proposed an incremental default risk charge (IDRC) in 2005 that would be calculated with a 99.9% confidence level and a one-year time horizon for instruments in the trading book that were sensitive to default risk.
- D. The comprehensive risk measure (CRM) is designed to take account of risks in asset-backed securities.

Q-78. The following formula defines the capital requirement (c) under the internal models

approach to the calculation of market risk under Basel III:

$$C = \max \{ \text{VaR}_{t-1}, m_c \times \text{VaR}_{\text{avg}} \} + \max \{ s\text{VaR}_{t-1}, m_s \times s\text{VaR}_{\text{avg}} \}$$

About this calculation, each of the following is true EXCEPT which is false?

- A. The first term is the higher of (i) the previous day's VaR and (ii) an average of the daily VaR measures on each of the preceding sixty business days, multiplied by a multiplication factor.
- B. The second term is the higher of (i) the latest available stressed VaR and (ii) an average of the stressed VaR numbers over the preceding sixty business days, multiplied by a multiplication factor.
- C. The multiplication factors m_c and m_s will be set by individual supervisory authorities but subject to an absolute minimum of three.
- D. The bank can choose to conduct an ex-post backtest on the stressed VaR only; if the test is successful, both multiplicative factors can be reduced to one.

Q-79. The capital conservation buffer:

- A. Will provide an extra 2.5% Common Equity Tier 1 capital buffer in times of stress.
- B. Will be used exclusively to protect banks from the losses garnered from OTC derivatives trading.
- C. Is required only for banks with inadequate liquidity coverage and net stable funding source ratios.
- D. Is covered in the increased Common Equity Tier 1 capital to risk-weighted assets ratio that will increase to 4.5% from the current 2% over the next few years.

Q-80. Which is true about the capital conservation buffer?

- A. When a bank's capital levels fall within this range, the bank can continue to conduct (operate) business.
- B. When a bank's capital levels fall within this range, the bank is constrained (restricted) with respect to dividends, share buybacks, and discretionary bonus payments to staff.
- C. When a bank's capital levels fall within this range, the bank is "severely restricted" with

respect to conducting business (operations).

- D. The bank can elect to draw down the buffer in normal times if competitive demands warrant, including the need to maintain market share.

Q-81. Capital conservation buffers have been established by the Basel Committee as part of measures designed to ensure that banks have enough capital to handle stress situations. Assuming no regulatory add-ons have been imposed, which of the following is correct?

- A. If the bank has 8% Common Equity Tier 1(CET1) capital with no Additional Tier 1 or Tier 2 capital. It would have zero conservation buffer and therefore be subject to a 100% constraint on capital distributions.
- B. If the bank has 8% CET1 with no Additional Tier1 or Tier 2 capital, it would satisfy the zero conservation buffer and therefore not be subjected to a constraint on capital distributions.
- C. If the bank has 7% CET1 with no Additional Tier1 or Tier 2 capital; it would have a 2.5% conservation buffer and therefore not be subjected to a constraint on capital distributions.
- D. If the bank has 9.5% CET1 with no Additional Tier 1 or Tier 2 capital, it would have a 2.5% conservation buffer and therefore not be subjected to a constraint on capital distributions.

Q-82. Among these two buffers, which does Basel 3 implement to reduce procyclicality and “promote the conservation of capital and the build-up of adequate buffers above the minimum that can be drawn down in periods of stress?”

- I. Basel 3 will phase-in a capital conservation buffer of 2.5% (of RWA) comprised of common equity Tier 1
- II. Basel 3 will phase-in a countercyclical buffer of between 0% and 2.5% (of RWA) to be determined by supervisors (national authorities)
- A. Neither I nor II.
- B. Only I but not II.

- C. Only II but not I.
- D. Both I and II.

Q-83. Which is true about the countercyclical conservation buffer?

- A. The countercyclical buffer is primarily a micro-prudential measure
- B. The countercyclical buffer can only be zero (0%) during the phase-in period, as eventually it achieves a constant of 2.5% regardless of environment
- C. Its primary goal is to avoid destabilizing losses subsequent to a period of excess credit growth
- D. A bank will be required to maintain this buffer if the bank falls under a jurisdiction identified and designated by the Basel Committee

Q-84. Which of the following statements is false regarding the leverage ratio? The leverage ratio:

- A. Acts as a supplementary measure to risk-based capital standards.
- B. Is defined as Tier 1 capital to on-and-off-balance sheet items and exposures.
- C. Allows banks to lend approximately 33 times their capital.
- D. Is risk-based.

Q-85. The CFO at a bank is preparing a report to the board of directors on its compliance with Basel requirements. The bank's average capital and total exposure for the most recent quarter is as follows:

REGULATORY CAPITAL	USD MILLION
Total Common Equity Tler 1 Capital	108
Additional Tler 1 Capital	28
Prior to regulatory adjustments	34
Regulatory adjustments	6
Total Tler 1 Capital	136
Tler 2 Capital	36

Prior to regulatory adjustments	45
Regulatory adjustments	9
Total Capital	172
Total Average Exposure	3678

Using the Basel III framework, which of the following is the best estimate of the bank's current leverage ratio?

- A. 2.94%
- B. 3.70%
- C. 4.68%
- D. 5.08%

Q-86. Each of the following is a characteristic of a high-quality liquid asset except for:

- A. Active and sizeable market with evidence of market breadth (price impact per unit of liquidity) and market depth (units of the asset that can be traded for a given price impact)
- B. High market concentration among a limited set group of buyers and sellers
- C. Low correlation with risky assets; i.e., not subject to wrong-way risk
- D. Asset class has shown historical tendency to be a "flight to quality" destination

Q-87. The liquidity requirement designed to improve bank resiliency to liquidity shocks over a one-year horizon is called the:

- A. Liquidity coverage ratio.
- B. Net stable funding ratio.
- C. Contractual maturity mismatch ratio.
- D. Available unencumbered assets ratio.

Q-88. According to the Basel Committee, "During the early liquidity phase of the financial crisis, many banks--despite adequate capital levels--still experienced difficulties because they did not manage their liquidity in a prudent manner. The crisis again drove home the importance of liquidity to the proper functioning of financial markets and the banking

sector. Prior to the crisis, asset markets were buoyant and funding was readily available at low cost. The rapid reversal in market conditions illustrated how quickly liquidity can evaporate and that illiquidity can last for an extended period of time...the Committee has further strengthened its liquidity framework by developing two minimum standards for funding liquidity.” Consider the following statements:

- I. The two mentioned standards - which aim to strengthen the liquidity framework - are the liquidity coverage ratio (LCR) and the net stable funding ratio (NSFR)
- II. The LCR tries to ensure short-term liquidity resilience (one month) while the NSFR promotes longer-term liquidity resilience (one year)
- III. The LCR anticipates an acute stress scenario by defining total net cash outflows under a stress scenario, while the NSFR does not explicitly simulate a stress scenario

Which of the above is (are) true?

- A. None are true
- B. I only
- C. II only
- D. All are true

Q-89. Which of the following statements is correct regarding capital requirements for insurance companies?

- A. Basel II includes the regulation of banks and insurance companies in the three pillars.
- B. The minimum capital requirement is likely to be higher than the solvency capital requirement for insurance companies.
- C. The repercussion for violating the solvency capital requirement is likely liquidation and the transfer of company insurance policies to another firm.
- D. The internal models approach to calculating the solvency capital requirement is similar to internal ratings based approach under Basel II in that the firm must calculate a VaR with a one-year time horizon.

Q-90. Which of the following statements is correct regarding the mechanics and motivations

of contingent convertible bonds (CoCos)?

- I. During normal financial periods, CoCos are debt and do not drag down return on equity.
 - II. During periods of financial stress, CoCos convert to equity, providing a cushion against loss, which helps prevent insolvency.
- A. I only.
 - B. II only.
 - C. Both I and II.
 - D. Neither I nor II.

Q-91. An operational risk manager is asked to report a bank's operational risk capital under the Standardized Measurement Approach (SMA) proposed by the Basel Committee in March 2016. The treasury department produces the following data for the bank, calculated according to the SMA guidelines:

- Business Indicator (BI): EUR 1,200 million
- Internal Loss Multiplier: 1

In addition, the manager uses the Business Indicator buckets in the Business Component presented in the table below:

Bucket	BI Range	BI Component
1	$\leq 1\text{billion}$	$12\% \times \text{BI}$
2	$1\text{ billion} < \text{BI} \leq 30\text{billion}$	$\text{EUR}120\text{million} + 15\%(\text{BI} - \text{EUR}1\text{billion})$
3	$> 30\text{billion}$	$\text{EUR}4.47\text{billion} + 18\%(\text{BI} - \text{EUR}30\text{billion})$

What is the correct operational risk capital that the bank should report under the SMA?

- A. EUR 120 million
- B. EUR 150 million
- C. EUR 158 million
- D. EUR 180 million

Q-92. The Basel III reforms restricted the use of internal model approaches for all of the following risk categories except:

- A. credit risk.
- B. systemic risk.
- C. operational risk.
- D. credit valuation risk.

Q-93. The Basel III reforms introduced a leverage ratio buffer for all:

- A. global banks.
- B. unregulated global banks.
- C. global systemically important banks.
- D. banks regulated by the U.S. Federal Reserve and the European Banking Authority.

Q-94. The new operational risk capital requirements under the Basel III reforms are determined by measures of the bank's:

- A. leverage and income.
- B. income and historical operational risk losses.
- C. income and expected operational risk losses.
- D. leverage and expected operational risk losses.

Q-95. A regulatory analyst at a large bank is preparing a report of the bank's credit risk capital for the current year and uses the Basel II IRB approach for making the calculation. As part of this process, the analyst identifies a portfolio of credit exposures of equal size that are held by borrowers with the same probability of default. The analyst has collected the following information about the portfolio:

What is the correct estimate of the Basel II credit risk capital that the bank should reserve for this portfolio?

Exposure at default	EUR 200 million
1-year expected loss on the portfolio	EUR 4.2 million

Expected recovery rate on a defaulted credit	30.00%
1-year portfolio default rate at the 95 th percentile	5.66%
1-year portfolio default rate at the 99.9 th percentile	9.87%

- A. EUR 3.72 million
- B. EUR 5.92 million
- C. EUR 9.62 million
- D. EUR 13.82 million

Q-96. After reviewing the results of a successful mid-year stress test performed by EveRate Bank (ERB), a risk analyst at a national financial supervisory authority prepares a summary report showing how the quality of the bank's overall risk management compares with that of its industry competitors. ERB determines its capital adequacy and planning based on an internal assessment of risks, asset-liability efficiency, capital return analyses, and regulatory capital requirements. In preparing the report, the analyst uses five competitors as a proxy for the industry benchmark. The analyst assumes that individual trading positions for the bank and its competitors are of equal size. ERB reports assets and liabilities at market value. Relevant information about the bank and its peers under normal market conditions are shown below:

Selected Data for EveRate Bank	
Item	Value
10-day ES of the trading book at the 99% confidence level	USD 7 million
Core Tier 1 capital	USD 2 billion
Total risk-weighted assets	USD 25 billion
Total leverage exposure	USD 31 billion
Number of trading positions	200
Average default probability per year	5%
Average Measures for Five Peer-Group Banks (Industry Benchmark)	
Metric	Value
10-day VaR of the trading book at the 99% confidence level	USD 7 million
Basel III Tier 1 leverage ratio	6%

Number of trading positions	300
Hazard rate per year	5%

The analyst assumes that daily portfolio returns for ERB and its competitors are normally distributed with a common mean of zero and are serially independent to each other.

Which of the following conclusions is most appropriate to make about how ERB's performance compares to the industry benchmark?

- A. ERB's market risk of its trading book is higher.
- B. ERB's liquidity trading risk is higher.
- C. ERB's credit quality is higher.
- D. ERB's Basel III Tier 1 leverage ratio is higher.

Q-97. An operational risk manager is asked to report a bank's operational risk capital under the Standardized Measurement Approach (SMA) proposed by the Basel Committee in March 2016. The treasury department produces the following data for the bank, calculated according to the SMA guidelines:

- Business Indicator (BI): EUR 800 million

In addition, the manager uses the Business Indicator buckets in the Business Component presented in the table below:

Bucket	BI Range	BI Component
1	EUR 0 to EUR 1 billion	$0.12 * BI$
2	EUR 1 billion to EUR 30 billion	$EUR\ 120\ million + 0.15(BI - EUR\ 1\ billion)$
3	Higher than EUR 30 billion	$EUR\ 4.47\ billion + 0.18(BI - EUR\ 30\ billion)$

What is the correct operational risk capital that the bank should report under the SMA?

- A. EUR 90 million
- B. EUR 96 million
- C. EUR 108 million
- D. EUR 120million

Q-98. The CRO of a large bank has asked a group of risk managers to prepare a plan to

implement the revisions to the Basel III guidelines finalized in December 2017. The bank expects to qualify as a global systemically important bank (G-SIB) by the time the revisions are implemented. Which of the following actions is most appropriate for the bank to take to comply with the revised Basel III guidelines?

- A. Introduce an internally created model to determine the bank's operational risk capital.
- B. Assign different credit risk weights for residential mortgages based on each mortgage's loan-to-value ratio.
- C. Apply the advanced internal rating-based approach for certain types of credit exposures, such as exposures to banks and large corporations.
- D. Decrease the bank's target level of its countercyclical buffer and its Basel III Tier 1 leverage ratio.

QUESTION 94 THROUGH 95 REFER TO THE FOLLOWING INFORMATION

The CRO of Bank LGX, a non-dividend-paying US-based bank is preparing a report to the board of directors on the bank's capital adequacy and planning. Bank LGX is subject to both the Basel framework and the US banking rules governing global systemically important banks (G-SIBs). The bank claims that it was in compliance with all the capital requirements in January 2016 as all Basel III phase-ins have already occurred. The CRO is conducting the analysis for January 2017 using selected and most recent annual performance data, which are shown in the table below:

Item	Value
Common equity Tier 1(CET1) capital	1,515
Preferred stock(noncumulative)	100
Tier 2 capital	827
Risk-weight assets	26,395
Total assets	42,828
Total exposure	47,460

The CRO also reports the minimum regulatory capital requirements under the revised capital framework as presented in the table below. The capital ratios also include the capital conservation buffer of 2.5% (phased-in at an annual increment of 0.75%, starting January 2016) and a G-SIB surcharge of 3.0% (phased-in at an annual increment of 0.625%, starting January 2016) of risk-weighted assets to be reached by January 2019

	January 2016 Minimum Ratio	January 2017 Minimum Ratio	January 2018 Minimum Ratio	January 2019 Minimum Ratio
Capital conservation buffer	0.625%	1.25%	1.875%	2.5%
G-SIB surcharge	0.75%	1.5%	2.25%	3.0%
CET 1 ratio	4.5%	5.25%	6.5%	10.0%
Tier 1 capital ratio	6.0%	6.75%	8.0%	11.5%
Total capital ratio	8.0%	8.75%	11.5%	13.5%
Leverage ratio	4.0%	4.0%	4.0%	4.0%

Q-99. Given the regulatory benchmarks and the bank's performance, which of the capital requirements does Bank LGX satisfy as of January 2017?

- A. CET1 capital ratio only

- B. Leverage ratio only
- C. Tier 1 capital ratio and Leverage ratio only
- D. Total capital ratio and CET1 capital ratio only

Q-100. In viewing the results of this capital analysis report and other considerations for Bank LGX's capital planning, which of the following conclusions is correct?

- A. The capital conservation buffer can be met by an increase in Tier 2 capital.
- B. If the exposure on derivative asset positions decreases, holding other factors constant, total capital ratio would decrease.
- C. The increase in the credit valuation adjustment (CVA) due to the bank's asset counterparty positions would tend to raise the bank's risk-weighted assets.
- D. If the bank raises additional CET 1 capital and invests the same amount in gold, Bank LGX's net stable funding ratio (NSFR) will not change.

QUESTION 96 THROUGH 99 REFER TO THE FOLLOWING INFORMATION

ABC Bank operates in a G-20 country that has mandated full compliance with the Basel III Accord.

ABC Bank's current assets and capital are provided in the table below. The values in the grey cells can be derived from the data provided.

Regulatory Capital	USD Millions
Total Common Equity Tier 1 Capital	74
Prior to regulatory adjustments	83
Regulatory adjustments	9
Additional Tier 1 Capital	13
Prior to regulatory adjustments	16
Regulatory adjustments	3
Tier 1 Capital	87
Tier 2 Capital	36
Prior to regulatory adjustments	45
Regulatory adjustments	9
Tier 3 Capital	8
Total Capital	
Risk-Weighted Assets	USD Millions
Total Risk-Weighted Assets for Credit Risk	889
RWA for Credit Risk – Basel III IRB Approach	809
CVA Capital Charge	80
Total Capital Charge for Market Risk	26
Total Capital Charge for operational Risk	20
Other Pillar 1 Capital Requirements	0
Total Risk-Weighted Assets	
Capital Ratios	
Common Equity Tier 1 Capital Ratio	
Tier 1 Capital Ratio	
Total Capital Ratio	
Total Exposure for the Calculation of the Leverage Ratio	2,519
Leverage Ratio	

Q-101. An increase in which of the following would increase ABC Bank's Common Equity Tier 1 capital under the Basel III framework?

- A. Defined benefit pension fund liabilities.

- B. Investment in its own common shares.
- C. Unrealized loss on available for sale items.
- D. Goodwill and other intangibles.

Q-102. With respect to the Basel III accord, what is ABC Bank's current ratio of total regulatory capital to risk-weighted assets?

- A. 5.94%
- B. 8.40%
- C. 8.95%
- D. 13.16%

Q-103. With respect to the Basel III minimum capital requirements, without considering any buffers, which of the following statements about ABC Bank's current capital and assets is correct?

- A. ABC Bank currently satisfies the minimum requirement for both the Common Equity Tier 1 capital ratio and the Tier 1 capital ratio.
- B. ABC Bank currently satisfies the minimum requirement for Common Equity Tier 1 capital ratio, but not the Tier 1 capital ratio.
- C. ABC Bank currently satisfies the minimum requirement for Tier 1 capital ratio, but not the Common Equity Tier 1 capital ratio.
- D. ABC Bank currently does not satisfy the minimum requirement for either the Common Equity Tier 1 capital ratio or the Tier 1 capital ratio.

Q-104.  The financial crisis highlighted a number of shortcomings related to the use of internally modelled approaches for regulatory capital, including the IRB approaches to credit risk. These shortcomings include the excessive complexity of the IRB approaches, the lack of comparability in banks' internally modelled IRB capital requirements and the lack of robustness in modelling certain asset classes. To address these shortcomings, the Committee has made several revisions to the IRB approaches. Which of the following

approach under Basel III can be applied for equity, large corporation and financial institution simultaneously?

- A. IRB
- B. F-IRB
- C. SA
- D. BIA

Q-105. With respect to the Basel III accord, what is ABC Bank's current leverage ratio and does it satisfy the minimum requirement?

- A. 2.94%; yes, this satisfies the minimum required leverage ratio.
- B. 2.94%; no, this does not satisfy the minimum required leverage ratio.
- C. 3.45%; yes, this satisfies the minimum required leverage ratio.
- D. 3.45%; no, this does not satisfy the minimum required leverage ratio.

Solution

Q-1. Solution: C

The definition of Business Disruption and System Failures is Losses arising from disruption of business or system failures; the definition of Clients, Products, and Business Practices is losses arising from an unintentional or negligent failure to meet a professional obligation to specific clients (including fiduciary and suitability requirements), or from the nature or design of a product; the definition of Execution, Delivery, and Process Management is Losses from failed transaction processing or process management, from relations with trade counterparties and vendors; the definition of Employment Practices and Workplace Safety is Losses arising from acts inconsistent with employment, health or safety laws or agreements, from payment of personal injury claims, or from diversity or discrimination events

Q-2. Solution: D

D is correct. The board of directors for banks have responsibility for: 1) approval of risk management policies and procedures, 2) ensuring that operating managers have the required technical capabilities, 3) evaluating performance of risk management activities, and 4) maintain oversight of risk management activities.

Q-3. Solution: B

Sound operational risk governance, according to Basel, relies on three lines of defense: (i) first lines of defense – business line management, which is responsible for identifying and managing the risks inherent in the products, activities, processes and systems for which it is accountable; (ii) second line of defense – an independent corporate operational risk management function, generally complementing the business lines' operational risk management activities; (iii) third line of defense – an independent review – review and audit of the bank's operational risk management controls, processes and systems.

Basel II and Basel III define operational risk (inclusive of technological risk) as the risk of direct or indirect loss resulting from inadequate or failed internal processes, people and systems or from external events. Although a number of financial institutions add reputation risk and strategic risk (e.g., due to a failed merger) as part of a broadened definition of operational risk, they are not

within the scope of definition by Basel II/III.

Q-4. Solution: A

Management has the ability to influence the correlations between these risks by changing the asset/Liability mix, so management decision-making is indeed quite relevant.

Q-5. Solution: B

Statements (A), (C), and (D) are consistent with industry data. However, with operational risk, there tends to be large numbers of small losses and a small number of large losses, so the distribution is asymmetric (and fat-tailed)

Q-6. Solution: A

Manual collection from peer data is other source to solve the problem of data lacking.

Q-7. Solution: D

D is correct. Operational resilience relates to the ability of firms and the financial sector as a whole to prevent, adapt, respond to, recover from, and learn from operational disruptions. Importantly, regulators view operational resilience from the perspective of “important business services” which would cause intolerable levels of harm to consumers or market integrity if they are disrupted. The firm’s trading platform is an example of an important business service and as such these actions should improve its resilience and help to comply with regulatory expectations.

A is incorrect. Increasing internal performance incentives does not necessarily increase the ability to deliver its processes reliably.

B is incorrect. While consolidating costs from multiple systems might be efficient, human resources and payroll are not customer facing systems and improvements to them would not benefit customers or market integrity.

C is incorrect. Reserving additional economic capital for operational risk, while potentially beneficial for risk management purposes, does not directly improve the ability of any of the firm’s important business services to become more resilient to potential disruptions.

Q-8. Solution: C

C is correct. The board of directors of a bank should ensure that the bank's framework is subject to independent review by audit or other appropriately trained parties.

A is incorrect. Banks should view risk transfer tools, such as outsourcing and insurance, as complementary to a process of thorough internal operational risk controls. Outsourcing should not be viewed as a replacement for internal controls or to relieve management of the responsibility to manage operational risk. In addition, outsourcing can actually introduce operational risk to the bank.

B is incorrect. The new Basel recommendation of the standardized approach for all banks phased out the earlier Advanced Measurement Approaches, in which some large banks developed internal models to determine their regulatory capital.

D is incorrect. This refers to the business lines, which should serve as the primary owners of the risk taken within their business lines, and not the risk management function. Instead, the risk management function should develop and maintain policies to manage operational risk, review the business lines' risk management activity and have the power to challenge the relevance and consistency of the business unit's implementation of risk management controls. These responsibilities do not make the risk management function the primary owner of the risk.

Q-9. Solution: C

C is correct. Regulators generally only require the recording of losses directly identifiable as negative financial effects of operational incidents. Known legal costs incurred as a result of the loss are an example of direct financial losses and should be included as part of the report.

A is incorrect. The payment from the insurance company is considered a recovery and should be included in the report.

B is incorrect. Only direct costs are required to be reported to the regulator. The foregone revenues due to the destroyed buildings cannot be directly quantified as a cost incurred by the bank and therefore are not required to be included in the report. Generally, estimates of foregone revenues due to operational loss events do not qualify to be included in loss reports because they are not direct costs or estimates such as a loss provision for expected costs in the future.

D is incorrect. Damage due to natural disasters is an operational risk, classified under the Damage to Physical Assets category, and should not be classified as market risk. Also, even in cases where operational risk is more directly related to market risk (such as an unintentional error in booking a position that results in a market risk loss), the “boundary event” between operational risk and market risk should be classified as an operational risk.

Q-10. Solution: B

B is correct. The Swiss cheese model describes the layering of multiple controls, in which control weaknesses in each layer appear like holes in a slice of Swiss cheese and act like their position constantly changes. In this model, all the control layers would have to fail simultaneously (i.e. the holes in all layers would need to be aligned) in order for an operational loss to result.

A is incorrect. The multiple control layers in a Swiss cheese framework imply that a failure of one control does not automatically result in a loss, since other control layers would need to fail too.

C is incorrect. Controls should be structured so that the probability of failure of different controls should be independent. While a decreasing probability of subsequent control failures would be beneficial, in practice achieving this might not be realistic.

D is incorrect. The controls are assumed to be independent, and given this structure in which multiple controls must fail for an operational loss to occur, the probability of both smaller and larger losses is likely to decrease after the model is implemented.

Q-11. Solution: C

C is correct. For a large bank, insurance is appropriate for exposures in which the operational risk is quite predictable and known in their distribution of likelihood and impacts, and where the potential risk is large enough to have a significant effect on the bank’s profit and loss profile (i.e. tail risk.)

Both cyber risk and business discontinuity are risks that are significant enough but yet well understood enough to be transferred through insurance.

A is incorrect. Captive insurance or self-insurance are most appropriate for large organizations to use in transferring smaller exposures below a certain threshold, not for transferring tail risk.

B is incorrect. Insurance recoveries (not premiums) can be deducted from gross losses to calculate

net losses, which also influences the capital calculation and decreases required capital but not on a one-to-one basis. Premiums are not deducted.

D is incorrect. For a large traditional bank, core credit-related activities are not typically outsourced. It is also inappropriate for banks to outsource its account review process as it must perform due diligence on all potential customers to effectively manage fraud risk as well as ML/FT risk. Activities that are typically outsourced include non-core activities such as IT server management, cloud computing, or call centers.

Q-12. Solution : B

B is correct. Business line managers are part of the first line of defense. As the primary risk owners, they should have the authority to expose the bank to risks within its risk appetite limits.

A is incorrect. The risk management function, which is part of the second line of defense, should have this responsibility. The third line of defense is an independent review of the firm's risk management framework by internal or external auditors.

C is incorrect (cf: Foundations of Risk Management, chapter 8). As discussed in the Foundations section, risk culture indicators are effective metrics to include as part of an ERM program to track the current state of and trends in risk culture. However, they can not accurately quantify losses due to culture failings.

D is incorrect. The third line of defense, not the second line of defense, is responsible for performing an independent review of the design and effectiveness of the firm's risk management framework. This review should be performed by internal or external auditors who are independent of the risk management function or the firm's senior management.

Q-13. Solution: A

A is correct. The text specifically mentions the fact that all exchange rates cannot depreciate all at once as an example of coherence. As such, the auditor should be concerned with the assumption that USD, EUR, and JPY all depreciate at the same time, as these are three major world currencies that together are responsible for a large proportion of foreign exchange.

B is incorrect. Correlations between different investment assets typically increase in stress

situations. The impact on correlation between these two should be even stronger, as the period of stress should reduce the demand for the industrial commodities so they should become more positively correlated with equities (as both drop in price.)

C is incorrect. This should not provide a significant source of concern as this is what we would expect from a “flight to safety” phenomenon, in which capital flows from risky assets to a small number of very-low risk assets such as short-term AAA-rated sovereign debt.

D is incorrect. This should not provide a significant source of concern as this is exactly what we would expect monetary authorities to do when facing higher consumer prices.

Q-14. Solution: C

C is correct. As per the text, “In addition, a capital policy should include governance and escalation protocols that are clear, credible, and actionable in the event an actual or projected capital ratio target is breached.”

A is incorrect. The board of directors should have the responsibility for challenging assumptions in capital models and providing final approval for them. Business unit managers should not have the right to provide final approval for the plan for their business unit, as this is decided by the economic capital group and approved by the board.

B is incorrect. This method is overly conservative as it ignores diversification benefits.

D is incorrect. The use of third-party vendor models is not discouraged, but guidelines for banks that use third-party scenarios are provided. Banks are encouraged to tailor these scenarios to their specific risk profiles and unique vulnerabilities, but the guidelines do not say that the use of third-party vendor scenarios be avoided.

Q-15. Solution: C

C is correct. The Basel three lines of defense model establishes the following lines of defense: In the first line of defense business line managers manage the risk of their business lines, in the second line of defense the corporate operational risk function (CORF) reviews the risk controls put in place by the first line of defense and establishes firm-wide risk management procedures, and in the third line of defense, an independent review (such as an internal auditor) reviews the effectiveness of

the risk controls in the first two lines of defense. C is correct, since as part of the second line of defense, the CORF should challenge inputs from business line managers.

A is incorrect, as internal audit is part of the third line of defense and the validation team is generally part of the corporate risk function as part of the second line of defense.

B is incorrect. Business line managers do not challenge the audit function as part of the first line; rather, they manage the risk of the business lines.

D is incorrect, as the CORF is the second line of defense.

Q-16. Solution: B

B is correct. This loss was caused by an error in transaction capture, which is classified as an operational loss in the Execution, Delivery and Process Management category.

A is incorrect. This is an operational risk in the Clients, Products and Business Practices classification related to disputes over performance of advisory activities.

C is incorrect. This is a loss in the Internal Fraud category.

D is incorrect. This is an example of a credit risk loss for Bank JKY, even though it resulted from an operational risk event at the counterparty. Also, monetary loss due to a cyber-attack should be classified by the client in the External Fraud category.

Q-17. Solution: B

B is correct. One of the steps in developing an operational resilience framework is to identify the firm's important business services, which for this firm would likely include the bill payment services and the international currency payment services. These important business services should all be customer-facing services. Once the important business services are identified, the firm must next map the dependencies between the services to identify potential vulnerabilities or areas where a disruption in one important business service can affect others.

A is incorrect. An impact tolerance should be developed for important business services.

C is incorrect. Regulatory expectations for operational resilience do not include the provision of additional capital. Also, current Basel regulatory guidelines for operational risk no longer include an approach that requires the calculation of a 99.9% VaR, this approach (the Advanced

Measurement Approach) was phased out.

D is incorrect. Operational resilience should be a multi-disciplinary approach. The resilience team is usually coordinated by the operational risk management function and includes representatives from functions such as business continuity management. Legal and compliance functions, while important, do not have as much of the relevant operational or process-related expertise that is important in a resilience context.

Q-18. Solution: D

D is correct. Banks with stronger practices will incorporate forward-looking and idiosyncratic factors into their stress scenarios.

A is incorrect. Operational risks typically have a low correlation with other market risk variables so assuming a zero correlation is conservative and an acceptable practice (see p. 276). “Most BHCs were not able to find meaningful correlation between macroeconomic variables and operational-risk loss severity”. Banks can provide correlation estimates between OpRisk and market risk variables with a proper defense, but assuming that op risk and market risk variables are strongly correlated is a cause for concern.

B is incorrect: This is a weak practice: Fed Capital Planning: “In general, it is a weaker practice to combine two different models, as it can introduce unexpected jumps in estimated losses over the planning horizon”. The paper also detailed some difficulties with roll rate models, which estimate the rate at which loans that are current or delinquent in a given quarter roll into delinquent or default status in the next period. By not using the roll-rate model to model the near-term quarters, this could lead to poor predictive power farther out.

C is incorrect. This is a weak practice as it ignores potential seasonal patterns. Rather, a preferred method would be to provide a careful estimate of the expected quarterly path of losses as well as net revenues and capital projections.

Q-19. Solution: A

A is correct. Internal training programs are examples of directive controls. Directive controls describe all the policies, procedures, and rules to help execute a process as well as mitigate the risk

of that process.

B is incorrect. This is a detective control.

C is incorrect. This could be either a preventive control, if the update was implemented in the absence of any cyber risk incident, or a corrective control if it was performed in response to an incident.

D is incorrect. This is an example of an access control, which is a preventive control.

Q-20. Solution: B

Preventive Control (MFA): Designed to prevent unauthorized access before an incident occurs.

Example: Multi-factor authentication reduces the likelihood of compromised credentials.

Detective Control (IDS): Identifies and detects ongoing or past incidents. Example: Intrusion detection systems monitor networks for suspicious activity.

Corrective Control (Incident response team): Addresses and corrects the impact of an incident after detection. Example: Restoring systems post-breach minimizes operational disruption.

Directive Control (Cybersecurity policy): Directs behavior to align with organizational goals or regulations. Example: Mandatory training ensures employees follow security protocols.

Q-21. Solution: D

D is correct. The Capital One loss was caused by a data breach in which a former cloud-services vendor employee exploited a weakness in a misconfigured web application firewall to gain access to the files stored in an Amazon Web Services (AWS) database. As a result, tens of thousands of customer Social Security numbers and account numbers were stolen, and Capital One was fined USD 80 million dollars for “failure to establish effective risk assessment processes prior to migrating significant information technology operations to the public cloud environment”.

A, B, and C are incorrect. These did not contribute to the loss.

Q-22. Solution: C

C is correct. The central operational risk function should be responsible for collecting information about specific operational risk exposures from the business units and aggregating this information

into a firm-wide risk report.

A is incorrect. Reports to the board of directors should contain a less extensive set and concentrate on those metrics that provide an overview of the firm's key risk exposures and are necessary for decision-making. Business line managers and other first-line risk managers should monitor a much more extensive set of indicators to detect potential adverse trends across a wide variety of metrics.

B is incorrect. The firm's executive committee should have this responsibility.

D is incorrect. Reporting operational risks using a forward-looking perspective is often done, especially in firms with a more mature operational risk management function. However, risk reporting does not have to be exclusively forward looking and many firms still use a largely backward-looking approach. Examples of a valuable backward-looking approach is performing trend or volatility analysis on the firm's operational losses in different categories over time.

Q-23. Solution: B

Based on bank operational loss data for 2014-2019, CPBP accounted for 52% of loss severity (very high loss severity), which was by far the greatest of the seven types. It was followed by EDPM, which accounted for 27% of loss severity (high loss severity).

Q-24. Solution: C

The statement C is incorrect because accurately modeling the tail of operational loss distributions requires more than just adding higher value losses to the dataset. It involves using a combination of methods, including extreme value theory and scenario data, to adequately capture the extreme skewness and the tail risk of operational losses. This comprehensive approach is necessary to address the full spectrum of potential losses, especially those of high severity but low frequency. Statements A, B, and D accurately reflect the considerations and methodologies described in the reading material regarding the selection and aggregation of Units of Measure (UoM) in the context of modeling operational risk.

Q-25. Solution: B

I. Incorrect. Even though this is a risk that can increase exposure to model risk, the policy itself is regarding compensation and not the model itself.

II. Correct. This assumption can lead to major error where market liquidity is limited.

Q-26. Solution: A

A sound development process will produce documented evidence in support of all model choices, including the overall theoretical construction, key assumptions, data, and specific mathematical calculations. They should require maintenance of detailed documentation of all aspects of the model risk management framework, including an inventory of models in use, results of the modeling and validation processes, and model issues and their resolution. Sensitivity analysis and other checks for robustness and stability should likewise be repeated periodically.

Q-27. Solution: B

Six common model errors include: (1) assuming constant volatility, (2) assuming a normal distribution of returns, (3) underestimating the number of risk factors, (4) assuming perfect capital markets, (5) assuming adequate liquidity, and (6) misapplying a model.

Q-28. Solution: D

Country risk refers to using a service provider based in a foreign country and subjecting the financial institution to potential economic and political risks in that country. Clearly, it is not a relevant risk arising from the breach of confidential customer data. Compliance risk is a possibility given the apparent lack of security controls of the service provider that resulted in the data breach. Operational risk is clearly a relevant risk to the financial institution here given the data breach caused by the service provider. Legal risk is clearly a relevant risk given that the customers affected by the data breach may sue the financial institution as a result of the breach.

Q-29. Solution: A

A is correct. Internal Controls: "For significant service provider relationships, financial institutions should assess the adequacy of the provider's control environment. Assessments should include reviewing available audits or reports such as the American Institute of Certified Public Accountants' Service Organization Control 2 report."

B is incorrect. The bank should review the vendor's incentive compensation structure and ensure

that the structure does not encourage vendor sales representatives to direct customers towards higher margin products without regard for the risk incurred. Compensating sales reps mostly with commissions would not be an appropriate structure.

C is incorrect. Outsourcing critical processes is not ruled out as a guideline, for example: A community banking organization may have critical business activities being outsourced, but the number may be few and to highly reputable service providers "(Larger) financial institutions may use hundreds or thousands of service providers for numerous business activities that have material risk..."

D is incorrect. The bank should monitor the vendor's contingency planning process and "assess the adequacy and effectiveness of a service provider's disaster recovery and business continuity plan and its alignment with its own plan".

Q-30. Solution: C

C is correct. Due diligence should be applied not only to customers but also to persons acting on their behalf and beneficial owners.

A is not correct. Banks should still conduct their own due diligence on the customer in this case, as the previous account manager may have already closed the account because of concerns of illicit activity.

B is not correct. Due diligence should specifically be increased for customers with higher risk profiles.

D is not correct. Banks should not open an account under any circumstances if the client insists on remaining anonymous. Confidential "numbered accounts" do not function as anonymous accounts and should be subject to the same customer due diligence standards as other accounts.

Q-31. Solution: A

Jack's identity must be established and verified before carrying out any transactions.

In regard to (B), (C) and (D), each is FALSE. Instead, the following are true statements:

Banks are allowed to serve high-risk customers, but "if a bank chooses to do business with a high-risk customer, more intensive ongoing monitoring of that customer's activity is needed."

Previous or preexisting bank accounts do not exempt this bank's ML/FT obligations:

"Though information about a customer's previous banking relationships may be helpful, the fact that a customer previously had accounts at another bank is not sufficient to classify the customer as low-risk or as well-identified. For example, the previous bank may have rejected the customer due to ML/FT concerns."

Politically exposed persons (PEP) are high-risk and require the attention deserved by high-risk customers, but a bank is not required to avoid them.

Q-32. Solution: A

Training that is ultimately effective requires staff to fully buy-in to the process and participate; the more interactive, the better. One of the principles of gamification is to make participation voluntary, not mandatory. In fact, one study at a major corporation showed very clear and strongly-positive results with regard to phishing links and emails as a result of incentivized training. Treating cyber security as a competitive game, with scores posted, for example, is not inappropriate.

Q-33. Solution: A

Any form of anomaly detection (e.g., anomaly detection algorithm, signature-based detection method, network behavior anomaly detection) is prone to false positives or false negatives. Establishing a baseline of normal activity and behavior would mitigate such errors.

In contrast, penetration testing involves mock attacks on an organization's IT system performed by a hired hacker to determine how likely and how severe a real attack could be. The nature of penetration testing does not directly subject it to false positives or false negatives.

Q-34. Solution: C

For choice (A), operational risk refers to the risk of loss from inadequate or failed processes, people or systems or from external events.

For choice (B), impact tolerances realistically need to be relevant to outsourced (third-party) service providers. The PRA expects firms to be able to remain within impact tolerances for important business services, irrespective of whether or not they use third parties in the delivery

of these services. This means that firms should effectively manage their use of third parties to ensure they can meet the required standard of operational resilience.

For choice (D), impact tolerances differ from risk appetites in that they assume a particular risk has crystallised instead of focusing on the likelihood and impact of operational risks occurring. Firms that are able to remain within their impact tolerances increase their capability to survive severe but plausible disruptions, but risk appetites are likely to be exceeded in these scenarios.

Q-35. Solution: C

C is correct. Internal and external auditors should both play an important role in independently evaluating the bank's risk controls, in order to ensure that they are effective in managing ML/FT risk, comply with the bank's policies and procedures for managing this risk, and also comply with regulatory requirements and supervisory guidance. This role is part of the third line of defense in managing ML/FT risk.

A is incorrect. The chief AML/CFT (anti-money laundering/counter-financial terrorism) officer should not have business line responsibilities to prevent potential conflicts of interest with the business lines from occurring. Also, the chief AML/CFT officer is part of the second line of defense, and day-to-day management of these risks within business lines should be performed by the business lines themselves as part of the first line of defense.

B is incorrect. The chief AML/CFT officer should have this responsibility.

D is incorrect. Screening of current and prospective employees should be part of the first line of defense, and should therefore be performed by staff and management of the business lines.

Q-36. Solution: B

B is correct. The bank was fined in March 2022 after admitting that from January 2016 through April 2021, it failed to implement and maintain an anti-money laundering program that met the minimum requirements of the Bank Secrecy Act. Several major deficiencies in the bank's processes were highlighted by regulators over this time period, but they were not addressed by the bank, and the bank's failure to correct these deficiencies resulted in the eventual fine.

A, C and D are incorrect. These scenarios are not related to this case study.

Q-37. Solution: D

D is correct. While this makes the process more resource intensive, it helps the bank respond more quickly to increasingly dynamic market conditions with more frequent recalibrations of the models.

A is incorrect. If some internal audit staff perform certain validation activities, they should not be involved in the assessment of the firm's overall model risk management framework.

B is incorrect. Ongoing monitoring is a core element of the validation process and should begin when the model is first implemented in production systems for actual business use. Backtesting results are typically not available for a significant period of time after the model is put into production so beginning the monitoring process after the backtest would be too late.

C is incorrect. Vendor models are often designed to provide a range of capabilities and therefore might need to be customized by a bank to meet its specific operating circumstances. Banks should always verify whether any inputs or assumptions provided by the vendor remain relevant for the bank and the current operating conditions, and should document any customization to these models that is made.

Q-38. Solution: D

D is correct. Sensitivity analysis and other checks for the robustness and stability of a model should be repeated periodically. This analysis can help assess the model's performance given a wide range of market conditions and identify those conditions (such as interest rate or financial market environments) in which the model might not perform as effectively and therefore its use should be constrained under these conditions.

A is incorrect. Even a fundamentally sound model producing accurate outputs consistent with the design objective of the model may exhibit high model risk if it is misapplied or misused. Applying a residential mortgage model to a commercial loan product would be example of misapplication.

B is incorrect. Best practices do not preclude the inclusion of qualitative or judgmental aspects of models. In some cases, banks may take statistical output from a model and modify it with judgmental or qualitative adjustments as part of their model development process. While such practices may be appropriate, banks should ensure that any such adjustments made as part of the

development process are conducted in an appropriate and systematic manner, and are well documented.

C is incorrect. Validation should be performed by parties who are independent of the model development team to avoid conflicts of interest.

Q-39. Solution: B

B is correct. There are two embedded options in non-maturity deposits. The first is the bank's option to change the interest rate and to determine what interest rate to charge its depositors. The second is the depositors' option to put the deposits back to the institution by withdrawing them. Given these characteristics, non-maturity deposits can be viewed as floating rate, puttable bonds.

A is incorrect. In addition, while banking institutions may change the offered deposit rates when market interest rates change, they do so with a lagged response, and by less than the full amount of the change in market rates.

C is incorrect. Mortgages will have negative convexity as their value increases less than an option-free bond because of the prepayment option.

D is incorrect. Interest rate risk and credit risk can interact, especially in stress situations, so the bank should incorporate a rule that links credit spread to changes in macroeconomic variables and interest rates.

Q-40. Solution: D

D is correct. The agreement should define specific events that constitute a default, and the bank should give the provider an opportunity to cure the default if one of these events occur instead of immediately terminating the contract.

A is incorrect. While the bank should audit the provider's processes, the bank shouldn't give the provider the right to audit its own processes.

B is incorrect. Neither of these actions are required to comply with best practices, although the bank should assess the adequacy of the provider's insurance policy.

C is incorrect. The provider should develop its own incentive compensation structure, and the bank should review it. The bank should ensure that sales reps do not receive an incentive compensation

structure which would encourage the bank or its customers to take too much risk, and a high proportion of variable compensation would be problematic in this respect.

Q-41. Solution: B

B is correct. Sharing of information and collaboration among banks depends on the financial industry's culture and level of trust among participants. Experience shows that a two-level information-sharing structure through which information would be first shared on the interpersonal level with a closer group and then be exchanged at the company level with a broader group of banks helps build trust into the system.

A is incorrect. Sharing of information among banks is one of the most widely observed practices across jurisdictions and a relatively wider range of information, such as knowledge about cyber threats / cyber intelligence is typically shared among banks.

C is incorrect. Sharing amongst regulators is one of the least observed practices and a majority of jurisdictions do not currently allow it.

D is incorrect. Banks typically do not share information about cyber-incidents with each other, but they do share this information with regulators at times when required by regulatory reporting practices.

Q-42. Solution: A

A is correct. As part of the first line of defense, business units must identify, assess, and control ML/FT risks, have written policies and procedures as well as employee training for managing ML/FT risk, and screen potential employees.

B is incorrect. Monitoring for ML/FT risk should cover all accounts and transactions.

C is incorrect. Politically exposed persons (PEP) pose higher ML/FT risk given the possibility that some wealth may have been obtained through corruption.

D is incorrect. A chief ML/FT officer should be appointed to lead the bank's management of ML/FT risk, and ideally ML/FT risk should be managed using a three lines of defense approach involving multiple business functions.

Q-43. Solution: C

The Dodd-Frank Act, while primarily aimed at reducing systemic risk following the 2008 financial crisis, also included significant provisions for consumer protection. One of the key components was the creation of the Consumer Financial Protection Bureau (CFPB), specifically designed to protect consumers in the financial sector. This makes option C incorrect as it suggests that Dodd-Frank did not include consumer protection measures. Options A, B, and D accurately reflect the consumer protection efforts within MiFID I and MiFID II, emphasizing transparency, product governance, and the suitability of financial products for clients.

Q-44. Solution: C

NIST CSF (Requirement 1): The NIST Cybersecurity Framework is voluntary and structured around five core functions: Identify, Protect, Detect, Respond, Recover . It provides best practices for managing cyber risks without mandating compliance, making it ideal for organizations seeking flexible guidance.

CIS Controls (Requirement 2): The CIS Critical Security Controls offer a prioritized list of 18 actionable controls (e.g., secure configurations, access management) to address common threats. These controls are regularly updated (e.g., Version 8 for cloud and remote work) and focus on practical implementation.

ISO/IEC 27001 (Requirement 3): ISO/IEC 27001 is an international certification standard requiring organizations to establish an ISMS and follow the PDCA cycle (Plan-Do-Check-Act) for continuous improvement. It emphasizes systematic risk assessment and compliance audits.

Q-45. Solution: B

	Initial	Operating Expenses↓	Revenue ↑	Return on EC↑	Interest Expense↓
Expected Revenue	16.00	16.00	19.2	16.00	16.00
Return on EC	0.94	0.94	0.94	1.88	0.94
Interest Expenses	4.00	4.00	4.00	4.00	2

Operating Expenses	2.00	1	2.00	2.00	2.00
Expected Loss	1.00	1.00	1.00	1.00	1.00
EC	47.00	47.00	47.00	47.00	47.00
RAROC	21.15%	23.28%	27.96%	23.15%	25.40%

Q-46. Solution: C

The cost of equity represents the minimum rate of return on equity required by shareholders.

Therefore, if RAROC is below the cost of equity, then there is no value being added.

Q-47. Solution: D

Expected revenue = \$2.0 billion loan portfolio × 6.0% = \$120.0 million

Expected losses = \$2.0 billion loan portfolio × 1.5% = \$30.0 million

Interest expense = \$2.0 billion borrowed funds × 2.0% = \$40.0 million

Operating cost = \$16.0 million (given as an assumption)

Economic capital = \$200.0 million = 10.0% × \$2.0 billion (given as an assumption)

Return on economic capital (EC) = \$2.0 million = \$200.0 EC × 1.0%

Tax rate = 0.25 (given as assumption)

Such that RAROC = $[(\$120.0 - 30.0 - 40.0 - 16.0 + 2.0) \times (1.0 - 0.25 \text{ tax rate})] / 200.0 = 13.50\%$

Adjusted RAROC = $\text{RAROC} - \beta(e) \times [R_{(m)} - R_f] = 13.50\% - 1.60 \times [8.0\% - 1.0\%] = 2.30\%$ and 2.30% is greater than the risk-free rate.

Q-48. Solution: A

$$r_{ce} = 2\% + 1.2 \times (7\% - 2\%) = 0.08, h_{AT} = (0.08 \times 500 + 0.04 \times 300) \div (300 + 500) = 0.065$$

Q-49. Solution: C

C is correct. Consider the basic adjusted RAROC (ARAROC) formula for a project:

$\text{ARAROC} = \text{RAROC} - \beta E \times (R_m - R_f)$, where;

βE = Beta of the equity of the firm; R_m = Expected market rate of return; R_f = Risk-free rate of interest

$\beta E \cdot (R_m - R_f)$ = Risk premium of the project.

ARAROC is simply “RAROC adjusted for the systematic riskiness of the returns”. ARAROC can be used in evaluating the project in the following way: If the project’s “RAROC less the project’s risk premium” is greater than the risk-free rate, then the firm’s shareholders are compensated for the non-diversifiable systematic risk they bear when investing in the activity, assuming the investors hold a well-diversified portfolio (i.e., the project adds value). That is, if the project’s ARAROC exceeds the risk-free rate, it should be accepted by the firm. Otherwise, if it is less than the risk-free rate, the project should be rejected.

Given RAROC = 15%, $\beta E = 1.8$, $R_m = 11\%$ and $R_f = 3\%$, one can compute $ARAROC = 0.15 - 1.8 \cdot (0.11 - 0.03) = 0.006 = 0.6\%$ and is less than $R_f = 3\%$. Thus, the project is rejected.

Q-50. Solution: B

B is correct.

The risk-adjusted after-tax return on capital (RAROC) is computed by:

$$RAROC = (\text{After-tax expected risk-adjusted net income}) / \text{Economic capital} \\ = (ER + ROEC - IC - OC - EL - \text{Taxes} \pm \text{Transfers}) / \text{Economic capital}$$

where,

$$\text{Economic capital} = \text{JPY } 4,200,000,000 \times 0.10 = \text{JPY } 420,000,000$$

$$ER = \text{expected revenue} = \text{JPY } 4,200,000,000 \times 0.032 = \text{JPY } 134,400,000$$

$$ROEC = \text{pre-tax return on invested economic capital}$$

$$= \text{Economic capital} \times 0.014 = \text{JPY } 420,000,000 \times 0.014 = \text{JPY } 5,880,000$$

$$IC = \text{interest expense} = \text{JPY } 4,200,000,000 \times 0.004 = \text{JPY } 16,800,000$$

$$OC = \text{Operating Cost} = \text{JPY } 4,200,000,000 \times 0.005 = \text{JPY } 21,000,000$$

$$EL = \text{expected loss} = \text{JPY } 4,200,000,000 \times 0.02 = \text{JPY } 84,000,000$$

$$\text{Taxes} = (\text{Revenue} + \text{Income} - \text{Interest} - \text{Operating Cost} - \text{Loss}) \cdot (\text{Tax rate})$$

$$= (134,400,000 + 5,880,000 - 16,800,000 - 21,000,000 - 84,000,000) \cdot (0.38)$$

$$= (\text{JPY } 18,480,000) \cdot (0.38) = \text{JPY } 7,022,400$$

$$\text{Therefore, numerator} = \text{JPY } 11,457,600 \text{ and so, } RAROC = 11,457,600 / 420,000,000 = 0.0273 = 2.73\%$$

A is incorrect. 0.27% is the result obtained when the economic capital is incorrectly taken to be JPY

4.2 billion instead of it being 10% of the loan amount.

C is incorrect. 4.40% is the result obtained when taxes are ignored.

D is incorrect. 10.73% is the result obtained when IC is added instead of subtracting in the numerator.

Q-51. Solution: C

One of the challenges of designing useful stress tests is coherence. The sensitivities and scenarios must be extreme but must also be reasonable or possible (i.e., coherent). Problems are inherently multi-factored making it more difficult to design a coherent stress test. Hook is dealing with the possibly incoherent results of the banks stress tests.

Q-52. Solution: D

A scenario is coherent when a change in one factor influences other factors in a logical manner. In this case, choice d is a coherent scenario since the Nordlandian economy depends heavily on exports of oil and natural gas, so therefore a sustained decrease in natural gas prices should lead to a decrease in stock prices as the domestic economy weakens. In stress testing banks, it is often challenging to develop scenarios where all factors behave coherently.

Q-53. Solution: A

A is correct. Five Spanish banks did not pass the EBA stress tests, which resulted in another series of stress tests in Spain that led to increased capital requirements at 11 Spanish banks.

B is incorrect. The SCAP was developed due to significant uncertainty about the strength of US banks coming out of the crisis, and 10 of the banks were required to raise a total of US 75 billion in capital.

C is incorrect. CCAR required only a stress scenario, while EBA included both a base case and a stress scenario. This is still incorrect today as EBA still requires a baseline scenario (CCAR does too now.)

D is incorrect. The SCAP test released loss rates by asset class including first lien mortgages, credit cards, and commercial real estate. This increased the disclosure dramatically over earlier US stress

tests.

Q-54. Solution: C

Poisson and Negative Binomial distributions are appropriate for loss frequency while Pareto, lognormal, Generalized Gamma, Transformed Beta, or Weibull distributions (fat-tailed) are generally used for loss severity

Q-55. Solution: A

It is the perfect correlation.

Q-56. Solution: A

False: The capital charges are added (CRC + MRC + ORC); Basel II gives no recognition for potential diversification benefits at this level of analysis.

In regard to (b), (c) and (d), each are true. In regard to (b), please note the “mutually reinforcing” aspect of the framework; and that Pillar One implies only minimum capital requirements.

Q-57. Solution: D

The bank’s long-term strategy and a mapping of new product development initiatives to planned target markets and customers.

The Third Pillar does not ask the bank to reveal such strategic insights (the nearest to this is likely indirect by way of remuneration-related disclosures)

Q-58. Solution: B

B is correct. Under the advanced internal ratings-based (Advanced IRB) approach banks supply their own estimates of probability of default (PD), loss given default (LGD) and exposure at default (EAD). Since LGD is dependent on recovery rates, this also implies that the recovery rates are modelled. On the other hand, under the foundation ratings based approach, they supply only PD, while LGD and EAD are set by the Basel Committee. So, C is incorrect.

A is incorrect. The standardized measurement approach for operational risk has eliminated the use

of internal models for modelling operational risk.

D is incorrect. Basel II.5 introduced a Comprehensive Risk Measure for credit sensitive instruments dependent on credit correlation. Banks could use their internal models to calculate the CRM, with supervisory approval, and these models included the estimation of recovery rates. However, the FRTB withdrew the use of the CRM for securitized products due to there being too much volatility between different banks' models in modeling securitizations. Banks are now required to use the standardized approach for these products.

Q-59. Solution: C

The supervisor's duties as part of the supervisory review process include:

Check compliance with Pillars I and III of Basel II Accord. Which would include credit risk mitigation and transparency requirements. Review internal control systems. Access internal capital management methods employed by the bank. So I and II are correct. Note that the foundation IRB approach. The bank provides its estimates for PD but uses supervisory estimates for LGD and EAD for corporate loans. So III is incorrect. Also, the impact of interest rate risk on the bank's capital position must be assessed by determining the impact of a 200 basis Point shock or its equivalent. Following the guiding principles of the Basel Committee, the current regulatory choice of a stress scenario focuses on parallel shifts in the yield curve of +/- 200 basis points.⁵⁰ The Committee acknowledges that the parallel shifts of +/- 200 basis points are relatively simplistic, but it argues that these shocks appear to adequately cover volatilities across G10 countries, even though the appropriateness of the proposed shock needs to be monitored on an ongoing basis, and recalibrated should the rate environment shift materially. So IV is also correct. Therefore, the correct Solution for this question is choice C.

Q-60. Solution: C

The loss year is excluded, so the charge is AVERAGE $(130,230) \times 15\% \text{ alpha} = \27 million

Q-61. Solution: A

Trading and sales: 18%

Corporate finance: 18%

Payment, settlement: 18%

Commercial banking: 15%

Agency services: 15%

Retail banking: 12%

Retail brokerage: 12%

Asset management: 12%

Q-62. Solution: B

As stated, the TRUE statement is: "This definition [of operational risk] includes legal risk, but excludes strategic and reputational risk."

In regard to (A), (C) and (D) each in true .

In regard to (C), please note that under BIA, "Fingers for any year in which annual gross income is negative or zero should be excluded from both the numerator and denominator when calculating the average."

Q-63. Solution: C

Both non- advanced approaches, including the standardized approach, multiply factors (alpha or beta) by gross income. Instead, the advanced measurement approaches (AMA) instead obtain a risk charge by internally estimating the unexpected loss (UL) at 99.9% confidence over one year; i.e., an OpRisk VaR approach.

Q-64. Solution: A

The standardized approach maps gross Income for business lines; the AMA allows banks to estimate unexpected operational losses with their own, presumably more accurate, internal models conditional on supervisor approval and meeting the associated quantitative and qualitative criteria.

Q-65. Solution: C

The internal models approach allows banks to use risk measures derived from their own internal

risk management models, subject to a set of qualitative conditions and quantitative standards. In terms of risk aggregation within market risk using the internal models approach, banks are explicitly allowed to recognize empirical correlations across broad market risk categories, and, thus, diversification benefits. The standardized approach for market risk, on the other hand, assigns capital separately to each of debt securities, equity securities, foreign exchange risk, commodity risk, and options without consideration for correlations between different types of instruments. Thus, C is correct and B is incorrect.

Also, A and D are incorrect because operational risk cannot be diversified.

Q-66. Solution: C

The revised market risk capital requirement (at 99.0% level) is:

$$\begin{aligned}\text{Market Risk Capital} &= \max(\text{VaR}_{t-1}, m_c \times \text{VaR}_{60\text{-day Avg}}) + \max(s\text{VaR}_{t-1}, m_s \times s\text{VaR}_{60\text{-day Avg}}) \\ &= \max(451, 3 \times 413) + \max(995, 3 \times 1,106) \\ &= \text{USD } 1,239 \text{ million} + \text{USD } 3,318 \\ &= \text{USD } 4,557 \text{ million}\end{aligned}$$

Q-67. Solution: A

The backtesting framework only includes three zones: green, yellow, and red. The plus factor determined from these zones is added to the multiplier floor of three.

Q-68. Solution: C

This is extremely false: Basel II relies heavily on external credit ratings and the Committee has a focus to REDUCE reliance on external ratings. For example, for claims on banks, the rules are somewhat complicated. National supervisors can choose to base capital requirements on the rating of the country in which the bank is incorporated. The risk weight assigned to the bank will be 20% if the country of incorporation has a rating between AAA and A A-, 50% if it is between A+ and A- 100% if it is between BBB+ and B-, 150% if it is below B-, and 100% if it is unrated.

In regard to (A), Basel III will add the Liquidity Coverage Ratio (LCR) and the Net Stable Funding Ratio (NSFR). In regard to (B), Basel III will phase-in the countercyclical buffer requirement. In regard

to (D), this is TRUE. The new leverage ratio (Tier 1/Total Exposure) will begin in 2013 as an additional measure.

Q-69. Solution: D

$$RWA = \$10,000,000 + \$300,000 \times 12.5 + \$500,000 \times 12.5 = \$20,000,000 ;$$

$$\text{Total capital ratio} = \$2 \text{ million} / \$20 \text{ million} = 10.0\%$$

Q-70. Solution: C

Minimum total Tier 1 + Tier 2 capital REMAINS at 8.0% through January 2019. Additional capital requirements are achieved by other means; e.g., Minimum Tier 1 phases up to 6.0%, capital conservation buffer (CCB), minimum common equity capital ratio. In regard to (A), (B), and (D), these are TRUE about Basel III.

Q-71. Solution: C

Basel 2.5 updated the incremental risk charge. (IRC)

In regard to (A), (B) and (D), each is true.

The primary, new elements in Basel III include: liquidity ratios (LCR and NSFR), the leverage ratio, CVA and wrong-way (counterparty) risk.

Q-72. Solution: D

Common Equity Tier 1 capital to risk-weighted assets must be 4.5% beginning January 1, 2015. Common stock plus retained earnings, not preferred stock, must make up the bulk of Common Equity Tier 1 capital. The requirements for Common Equity Tier 1 capital are the most-stringent, not the least-stringent. Investors of the common shares must have a residual claim to the assets.

Q-73. Solution: A

No, because Tier 1 Capital is not at least 8.5%

Basel III requires Core Tier 1 (Common Equity) of 7.0%, Tier 1 of 8.5%, and Total Capital of 10.5%:

Core Tier 1 (Common Equity) ratio of at least 7.0% = 4.5% + 2.5% Conservation Buffer. Thrift Bank holds exactly sufficient Common Equity: $2.8/40.0 = 7.0\%$.

Tier 1 (Common Equity + Additional Tier 1) ratio of at least 8.5% = 6.0% Tier 1 + 2.5% Conservation buffer. Thrift Bank only holds Tier 1: $3.0/40.0 = 7.5\%$

Total Capital ratio of 10.5% = 8.0% total capital + 2.5% Conversation Buffer. Thrift Bank holds Total Capital: $4.4/40.0 = 11.0\%$

Q-74. Solution: B

Confidence level is 99.9%. Securitizations are subject to the banking book capital requirements.

Q-75. Solution: B

The stressed value at risk should be calculated on a weekly basis. This measure is calculated by combining current portfolio performance data based on the 10-day, 99% confidence interval with firm's historical data from a significantly financially stressed period of the same portfolio.

Q-76. Solution: B

The capital charge is an arithmetic sum of charges across categories including interest rate risk, equity position risk, foreign exchange risk, commodities risk, and options risk. This is why a key criticism of the standardized approach is that it overcharges by ignoring the benefits of any diversification.

In regard to (A), this is false: Basel III abolished Tier 3 capital

In regard to (C), this is false: The more sophisticated banks with well-established risk management functions were allowed to use an "internal model-based approach" for setting market risk capital. Because it better reflected the benefits of diversification and led to lower capital requirements, the resultant capital must be at least 50% of the capital given by the standardized approach.

In regard to (D), this is false: The method ignores the benefits of any diversification.

Q-77. Solution: B

In 2008, the Basel Committee recognized that most of the losses in the credit market turmoil of 2007 and 2008 were from changes in credit ratings, widening of credit spreads, and loss of liquidity, rather than solely as a result of defaults. In regard to (A), (C) and (D), each is true. CRM is designed

to take account of instruments such as asset-backed securities (ABSs) and collateralized debt obligations (CDOs), which are sensitive to the correlation between the default risks of different assets.

Q-78. Solution: D

The ex-post backtest applies only to the VaR, not the stressed VaR. Further, the backtest increases (via a “plus”) the m_c factor by a factor of zero to 1.0; it does not reduce the minimum of 3.0. Essentially, a yellow-zone backtest result can imply a minimum factor, m_c , of at least four ($4.0=3.0+1.0$), or more if the supervisor requires.

To review, the capital requirement (c) is given by the following formula:

$$C = \max \{VaR_{t-1}, m_c \times VaR_{avg}\} + \max \{sVaR_{t-1}, m_s \times sVaR_{avg}\}$$

i.e., the sum of:

The higher of its previous day's value-at-risk number, $VaR(t-1)$; and an average of the daily value-at-risk measures on each of the preceding sixty business days, $VaR(avg)$, multiplied by a multiplication factor, m_c , plus

The higher of its latest available stressed-value-at-risk $sVaR(t-1)$; and an average of the stressed value-at-risk over the preceding sixty business days, $sVaR(avg)$, multiplied by a multiplication factor, m_s

In regard to (A), (B) and (C), each is true.

Q-79. Solution: A

The capital conservation buffer is intended to provide an extra cushion against loss in times of stress. It is 2.5% Common Equity Tier 1 capital to risk-weighted assets, which in effect increases the total Common Equity Tier 1 capital ratio to 7%.

Q-80. Solution: B

When a bank's capital levels fall within this range, the bank is constrained (restricted) with respect to dividends, share buybacks, and discretionary bonus payments to staff

In regard to (A), (C) and (D), each are false.

Q-81. Solution: A

Core tier 1	$4.5\% + 2.5\% = 7\%$
Tier 1	$6\% + 2.5\% = 8.5\%$
Total	$8\% + 2.5\% = 10.5\%$

Q-82. Solution: D

Both I and II.

Q-83. Solution: C

Its primary goal is to avoid destabilizing losses subsequent to a period of excess credit growth

Losses incurred in the banking sector can be extremely large when a downturn is preceded by a period of excess credit growth. These losses can destabilize the banking sector and spark a vicious circle, whereby problems in the financial system can contribute to a downturn in the real economy that then feeds back on to the banking sector. These interactions highlight the particular importance of the banking sector building up additional capital defenses in periods where the risks of system-wide stress are growing markedly."

In regard to (A), this buffer is quintessentially MACRO-prudential; e.g., The countercyclical buffer aims to ensure that banking sector capital requirements take account of the macro-financial environment in which banks operate."

In regard to (B), this is false as the countercyclical buffer is only meant to apply during excess credit regimes. "This requirement will be released when system-wide risk crystallizes or dissipates."

In regard to (D), the requirement for this buffer (from 0 to 2.5%) is delegated to the respective national authorities, not Basel.

Each Basel Committee member jurisdiction will identify an authority with the responsibility to make decisions on the size of the countercyclical capital buffer. If the relevant national authority judges a period of excess credit growth to be leading to the buildup of system-wide risk, they will consider, together with any other macroprudential tools at their disposal, putting in place a countercyclical buffer requirement. This will vary between zero and 2.5% of risk weighted assets,

depending on their judgment as to the extent of the build-up of system-wide risk.”

Q-84. Solution: D

The leverage ratio is simple and non-risk based and meant to act as a “backstop” measure of leverage.

Q-85. Solution: B

For Basel III purposes, the leverage ratio is $\text{Tier 1 Capital} / \text{Total Exposure} = 136 / 3,678 = 3.70\%$

Q-86. Solution: B

NOT HIGH, but rather: Low market concentration among a limited set group of buyers and sellers

“Characteristics of high-quality liquid assets

(a) Fundamental characteristics

Low credit and market risk: assets that are less risky tend to have higher liquidity. High credit standing of the issuer and a low degree of subordination increases an asset’s liquidity. Low duration, low volatility, low inflation risk and denomination in a convertible currency with low foreign exchange risk all enhance an asset’s liquidity.

Ease and certainty of valuation: an asset’s liquidity increases if market participants are more likely to agree on its valuation. The pricing formula of high-quality liquid asset must be easy to calculate and not depend on strong assumptions. The inputs into the pricing formula must also be publicly available. In practice, this should rule out the inclusion of most structured or exotic products.

Low correlation with risky assets: the stock of high-quality liquid assets should not be subject to wrong-way (highly correlated) risk. For example, assets issued by financial institutions are more likely to be illiquid in times of liquidity stress in the banking sector.

Listed on a developed and recognized exchange market: being listed increases an asset’s transparency.

(b) Market-related characteristics

Active and sizable market: the asset should have active outright sale or repurchase agreement (repo) markets at all times (which means having a large number of market participants and a high

trading volume). There should be historical evidence of market breadth (price impact per unit of liquidity) and market depth (units of the asset that can be traded for a given price impact).

Presence of committed market makers: quotes will most likely be available for buying and /or selling a high-quality liquid asset.

Low market concentration: a diverse group of buyers and sellers in an asset's market increases the reliability of its liquidity.

Flight to quality: historically, the market has shown tendencies to move into these types of assets in a systemic crisis."

Q-87. Solution: B

The net stable funding ratio is intended to promote medium-and long-term funding of the bank's activities. It is defined as the available amount of stable funding divided by the required amount of stable funding, and it must be greater than 100%.

Q-88. Solution: D

All are true

"These standards have been developed to achieve two separate but complementary objectives. The first objective is to promote short-term resilience of a bank's liquidity risk profile by ensuring that it has sufficient high quality liquid resources to survive an acute stress scenario lasting for one month. The Committee developed the Liquidity Coverage Ratio (LCR) to achieve this objective. The second objective is to promote resilience over a longer time horizon by creating additional incentives for a bank to fund its activities with more stable sources of one year and has been developed to provide a sustainable maturity structure of assets and liabilities."

Q-89. Solution: D

Solvency II, not Basel II, establishes capital requirements for insurance companies. The minimum capital requirement (MCR) is just that, a true floor and is thus likely to be lower than the solvency capital requirement (SCR). The repercussion for violating the MCR is likely the prohibition of taking new business and possible liquidation. The repercussion for violating the SCR is the requirement

of a plan to remedy the situation and bring the capital back to the required level. The internal models approach is similar to the internal ratings based approach under Basel II in that the insurance company must calculate a one-year VaR with a 99.5% confidence level (versus 99.9% confidence for banks under Basel II).

Q-90. Solution: C

Contingent convertible bonds (CoCos), unlike traditional convertible bonds, convert to equity when the company or bank is experiencing financial strains. During normal financial periods, the bonds are debt and thus do not drag down return on equity (ROE).

Q-91. Solution: B

B is correct. Under the revised Standardized Measurement Approach, operational risk capital is equal to the Business Indicator Component multiplied by the Internal Loss Multiplier.

The Business Indicator Component is determined by the Business Indicator (BI), which is made up of almost the same P&L items that are found in the composition of Gross Income (GI). The main difference relates to how the items are combined. The BI uses positive values of its components, thereby avoiding counterintuitive negative contributions from some of the bank's businesses to the capital charge (e.g. negative P&L on the trading book), which is possible under the GI. In addition, the BI includes income statement items related to activities that produce operational risk that are omitted (e.g. P&L on the banking book) or netted (e.g. fee expenses, other operating expenses) in the GI.

In this case, the BI is already given as EUR 1,200 million.

Therefore, with a BI of EUR 1,200 million falling into the BI range of Bucket 2, and given that the Internal Loss Multiplier is equal to 1, the calculation of the operational risk capital for the bank in Bucket 2 is calculated as follows:

SMA operational risk capital (Bucket 2) = BIC $\times$ 1 = EUR 120 million + 0.15(BI – EUR 1 billion) = EUR 120 million + 0.15(EUR 1,200 million – EUR 1,000 million) = EUR 150 million.

Q-92. Solution: B

The use of internal model approaches are restricted for credit risk, operational risk, and credit valuation risk.

Q-93. Solution: C

The Basel III reforms revise the leverage ratio framework by adding a leverage ratio buffer for G-SIBs that must be met with Tier 1 capital.

Q-94. Solution: B

Operational risk capital requirements are determined by measures of the bank's income and historical operational risk losses, both of which are assumed to be positively correlated to future operational risk.

Q-95. Solution: C

C is correct. The correct Basel II IRB credit risk capital for the portfolio is equal to: (Exposure at Default * Loss given Default * $DR_{99.9}$) – Expected Loss

where $DR_{99.9}$ is equal to the worst-case default rate for the portfolio of assets calculated using the Vasicek model. For retail credit cards, this calculation assumes a correlation of 0.04 between defaults on individual exposures, which is specified by the Basel committee. The loss given default is equal to 1 minus the recovery rate, so it is 70%. Therefore, the correct capital is $(200 * 70\% * 9.87\%) - 4.2$, or EUR 9.62 million.

Q-96. Solution: D

D is correct. The Basel III Tier 1 leverage ratio = (Core Tier 1 capital)/(Leverage exposure) = $2/31 = 6.45\%$, which is higher than the average 6% for the industry peers. A higher leverage ratio for capital requirement is better.

A is incorrect. Under the distributional assumption, VaR at a 99% confidence level is approximately equal to ES at a 97.5% confidence level. ES is the average observation of losses that exceed the VaR. Therefore, if Bank ERB's 99% ES is equal to the industry benchmark's 99% VaR, it can be argued that Bank ERB's market risk is lower, not higher, than the benchmark since all of the benchmark

observations above the VaR will need to be higher than USD 7 million. Also, the portfolio return distributions of ERB and the peer firms may be different.

B is incorrect. There is not enough information to determine the levels of liquidity trading risk (or, e.g., Liquidity adjusted VaR) for both the bank and the benchmark. ERB's liquidity trading risk is not necessarily higher just because it has a smaller number of trading positions. A higher number of trading positions may induce a diversification effect, which has no effect on liquidity trading risk.

C is incorrect. Assuming a constant hazard rate, and using the hazard rate process, the average default probability for the industry benchmark = $PD = 1 - \exp(-\lambda \cdot t) = 1 - \exp(-0.05 \cdot 1) = 4.88\%$, where $t = 1$ year. Since ERB's PD (= 5%) is higher, its credit quality is lower.

Q-97. Solution: B

B is correct. Under the revised Standardized Measurement Approach, operational risk capital is equal to the Business Indicator Component multiplied by the Internal Loss Multiplier.

In this case, the BI is already given as EUR 800million.

Therefore, with a BI of EUR 800 million falling into the BI range of Bucket 1. When $BI < 1$ billion, the ILM is equal to 1. So, the calculation of the operational risk capital for the bank in Bucket 1 is calculated as follows:

$BIC(\text{Bucket 1}) = BI \cdot 0.12 = 96 \text{ million}$,

$SMA \text{ operational risk capital (Bucket 1)} = BIC \cdot ILM = 96 \text{ million} \cdot 1 = 96 \text{ million}$

Q-98. Solution: B

B is correct. One of the key changes in the December 2017 Basel reforms was the introduction of a set of variable risk weights for residential mortgages which varied based on the risk of the mortgage. This risk is reflected mainly in the loan-to-value (LTV) ratio with higher LTV mortgages receiving higher risk weights. This replaces an old flat rate system.

A is incorrect. The Basel reforms replaced the advanced measurement approach for operational risk, which required the use of internal models, with a standardized approach to be used by all banks. The new standardized approach does not allow internal modeling.

C is incorrect. The new reforms eliminated the use of the IRB approach for these asset classes.

D is incorrect. The 2017 reforms introduced a leverage ratio buffer that requires G-SIBs to have a higher minimum Tier 1 leverage ratio. Therefore, if anything, the target Tier 1 leverage ratio should increase. Also, the countercyclical buffer requirements were not changed as part of this reform.

Q-99. Solution: D

The bank's CET1 capital ratio = (CET 1 capital)/(risk-weighted assets) = $(1,515/26,395) = 5.74\%$. This ratio meets and exceeds the 5.25% minimum CET1 capital requirement;

The bank's leverage ratio = (Tier 1 capital)/(Exposure) = $(1,515 + 100)/(47,460) = 3.40\%$. This ratio does not meet the 4.0% minimum leverage ratio requirement;

The bank's Tier 1 capital ratio = (Tier 1 capital)/(risk-weighted assets) = $(1,515 + 100)/26,395 = 6.12\%$. This ratio does not meet the 6.75% minimum Tier 1 capital requirement;

The bank's Total capital ratio = (Total capital)/(risk-weighted assets) = $(1,515 + 100 + 827)/26,395 = 9.25\%$. This ratio meets and exceeds the 8.75% minimum Total capital requirement.

Q-100. Solution: C

C is correct. Increasing CVA charge increases the amount of risk-weighted assets.

A is incorrect. According to Basel, the conservation buffer can only be met by additional CET 1 capital.

B is incorrect. Derivative exposure (as well as other off-balance sheet items) are part of the total exposure. As exposure declines, Total capital ratio increases (assuming no change in Total capital).

D is incorrect. The NSFR = (amount of stable funding)/(required amount of stable funding). CET 1 capital, which goes to the numerator, has a weight of 100%. Gold, which goes to the denominator, has a weight of 50%. Thus, the increase to the numerator and denominator will not be exactly the same, so the NSFR changes.

Q-101. Solution: B

Tier 1: Predominant form must be common shares and retained earnings

- Common shares issued by the bank that meet the criteria for classification as common shares for regulatory purposes;

- Stock surplus (share premium) resulting from the issue of instruments included Common Equity Tier 1;
- Retained earnings;
- Accumulated other comprehensive income and other disclosed reserves;
- Minority interest
- Regulatory adjustments applied in the calculation of Common Equity Tier 1.

Q-102. Solution: B

Regulatory Capital = 87 + 36 = 123

Risk-weighted assets = $(26 + 20)/8\% + 889 = 575 + 889 = 1,464$

The ratio of total regulatory capital to risk-weighted assets = $123/1464 = 8.40\%$

Q-103. Solution: B

Basel III capital requirement:

- Common Equity Tier 1 must be at least 4.5% of risk-weighted assets at all times.
- Tier 1 Capital must be at least 6.0% of risk-weighted assets at all times.
- Total Capital (Tier 1 Capital plus Tier 2 Capital) must be at least 8.0% of risk-weighted assets at all times.

Common Equity Tier 1 capital charge = $74/1,464 = 5.05\%$

Tier 1 Capital = $87/1,464 = 5.94\%$

Q-104. Solution: C

Revised Scope of IRB Approaches for Asset Classes is as follow,

Portfolio/ Exposure	Basel II: Available Approaches	Basel III: Available Approaches
Large and mid-sized corporates (consolidated revenues > €500m)	A-IRB, F-IRB, SA	F-IRB, SA
Banks and other financial institutions	A-IRB, F-IRB, SA	F-IRB, SA
Equities	Various IRB approaches	SA
Specialised lending ³	A-IRB, F-IRB, slotting, SA	A-IRB, F-IRB, slotting, SA

Q-105. Solution: C

Leverage ratio = Capital / Total Exposure

Capital is Tier 1 capital

Leverage = $87/2,519 = 3.45\%$

The Committee will test a minimum Tier 1 leverage ratio of 3% during the parallel run period from 1 January 2013 to 1 January 2017.